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MSc in Computer Science — Multi Agent Systems Lab

Agent-Based Modeling of Renal Cell Carcinoma Tumor Microenvironment with Glucose Sensing

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Abstract

Renal cell carcinoma (RCC) accounts for approximately 400,000 new diagnoses annually worldwide, with metastatic disease carrying a five-year survival rate of roughly 15%. Understanding why individual patients respond differently to immunotherapy and targeted therapy remains a key challenge in clinical oncology.

This work presents an agent-based model (ABM) of the RCC tumor microenvironment comprising 18 distinct agent types—including tumor cells, six T cell subtypes, macrophages, dendritic cells, NK cells, blood vessels, and sex hormone agents—interacting on a three-dimensional discrete grid. Each tumor cell carries a 16-gene DNA profile that governs proliferation, immune evasion, and treatment sensitivity through codon-level mutation mechanics. A continuous glucose concentration field, governed by Fick's diffusion law, overlays the agent grid, coupling metabolic competition (the Warburg effect) to both tumor growth and immune effector function.

The model was ported from Mesa to Repast4Py to enable MPI-based parallel execution and integrates PD-1/PD-L1 checkpoint interactions, ICI and TKI treatment mechanisms, and sex hormone modulation of immune responses across 17 effect channels. The 87 tunable weight parameters are calibrated against ARON-1 clinical trial data using Bayesian optimization with Optuna.

A 160-simulation multi-seed validation study reveals sex-stratified treatment responses: female virtual patients receiving combination ICI+TKI therapy achieve the highest observed survival rate (65%), consistent with emerging clinical evidence for sex-dependent treatment efficacy. Statistical analysis using Mann-Whitney U tests characterizes the stochastic variability inherent in agent-based tumor–immune dynamics.

These results demonstrate that mechanistic agent-based models incorporating metabolic, genetic, and hormonal subsystems can generate clinically relevant predictions when calibrated against trial data, contributing to ongoing efforts in computational precision oncology.

Keyword List

Agent-based modeling, renal cell carcinoma, tumor microenvironment, glucose metabolism, Warburg effect, immune checkpoint inhibitors, tyrosine kinase inhibitors, sex hormones, Repast4Py, Bayesian optimization, ARON-1

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Table of Acronyms

Acronym	Expanded Version
ABM	Agent-Based Model
BMI	Body Mass Index
ccRCC	Clear Cell Renal Cell Carcinoma
CTL	Cytotoxic T Lymphocyte
DAMPs	Damage-Associated Molecular Patterns
DC	Dendritic Cell
HIF	Hypoxia-Inducible Factor
ICI	Immune Checkpoint Inhibitor
IMDC	International Metastatic RCC Database Consortium
MPI	Message Passing Interface
NK	Natural Killer (cell)
ODD	Overview, Design concepts, and Details (protocol)
OS	Overall Survival
PD-1	Programmed Cell Death Protein 1
PD-L1	Programmed Death-Ligand 1
pDC	Plasmacytoid Dendritic Cell
RCC	Renal Cell Carcinoma
TCR	T Cell Receptor
TKI	Tyrosine Kinase Inhibitor
TME	Tumor Microenvironment
TPE	Tree-structured Parzen Estimator
VEGF	Vascular Endothelial Growth Factor
VHL	von Hippel–Lindau

1. Introduction

> All models are wrong, but some are useful.
>
> — George E.P. Box

1.1 Renal Cell Carcinoma: Clinical Significance

Renal cell carcinoma (RCC) affects 400,000 people annually worldwide, with metastatic disease carrying a five-year survival rate of approximately 15% [27]. Clear cell RCC (ccRCC) constitutes 70–80% of cases and is characterized by *von Hippel–Lindau* (*VHL*) mutations leading to vascular endothelial growth factor (VEGF) overexpression—making it both a target for anti-angiogenic therapies and one of the tumors most responsive to immunotherapy due to its high immunogenicity [23]. Despite the success of checkpoint immunotherapy in reshaping the treatment landscape, approximately 40% of advanced RCC patients do not respond to first-line treatment, and there are currently no validated biomarkers that reliably predict which patients will benefit from immune checkpoint inhibitors (ICI) versus tyrosine kinase inhibitors (TKI) versus combination therapy—underscoring an urgent unmet need for predictive tools that can stratify patients and guide therapeutic decisions. Figure 1.1 summarizes the global incidence rates, sex distribution, and stage-dependent survival patterns that motivate the sex-stratified approach adopted in this work.

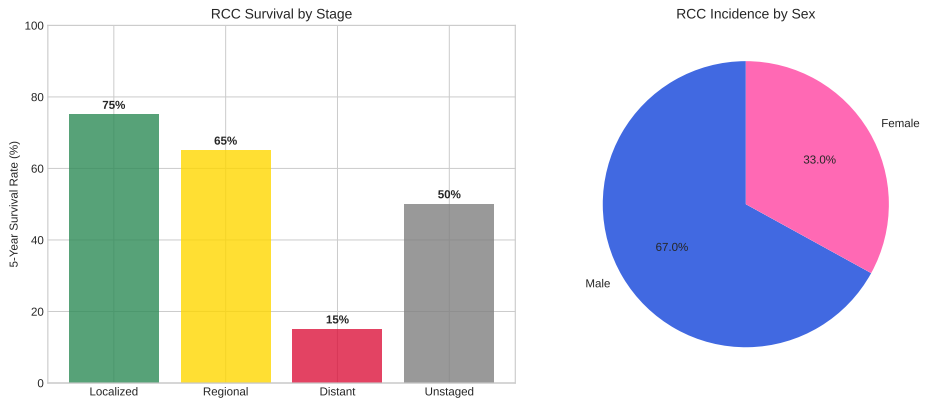

Figure 1.1: Renal cell carcinoma epidemiology data showing global incidence rates, sex distribution, and stage-dependent survival curves. The pronounced male predominance (2:1 ratio) and stage-dependent survival patterns motivate the sex-stratified modeling approach and clinical calibration used in this work.

1.2 The Tumor Microenvironment and Sex Differences

The tumor microenvironment (TME) is a complex, spatially heterogeneous ecosystem encompassing tumor cells, immune cells (cytotoxic T lymphocytes (CTLs), Th1/Th2, Tregs, DCs, M1/M2 macrophages, NK cells), vasculature, and signaling molecules. The bidirectional interactions between these components create nonlinear dynamics where immune cells can kill tumors but tumors can also co-opt immune responses to establish immunosuppressive niches [12].

Critically, the *spatial organization* of these cellular populations—not merely their relative abundances—determines clinical outcomes. Tumors classified as "immune-inflamed," in which CTLs infiltrate deep into the tumor parenchyma, respond markedly better to checkpoint immunotherapy than "immune-excluded" tumors, where effector cells accumulate at the tumor periphery but fail to penetrate the core, or "immune-desert" tumors, which lack significant immune infiltration altogether [6]. This spatial heterogeneity arises from physical barriers such as aberrant vasculature and dense extracellular matrix, as well as from chemokine gradients and metabolic competition that shape immune cell trafficking. Traditional ordinary differential equation (ODE) models, which represent cell populations as well-mixed compartments, cannot capture these spatially mediated phenomena: they cannot distinguish between a thousand T cells uniformly distributed throughout a tumor and a thousand T cells trapped at its margin—two configurations with profoundly different therapeutic implications. Agent-based models, by contrast, represent each cell as a discrete entity occupying a specific spatial position, making them a natural formalism for studying how tumor topology governs immune access and treatment efficacy.

RCC exhibits a pronounced sex bias, with males showing approximately twice the incidence of females across virtually all populations studied [31]. This disparity extends beyond incidence: male patients tend to present with higher-stage disease and experience worse stage-for-stage survival, suggesting that sex-linked biological factors influence progression and treatment response. Estrogen shows biphasic effects on immunity—enhancing innate and adaptive responses at physiological concentrations but promoting immunosuppression at higher levels—while testosterone generally suppresses immune responses and promotes immune evasion. Beyond hormonal effects, X-chromosome–linked immune genes (including *TLR7*, *FOXP3*, and *CD40L*) may contribute to stronger baseline immune surveillance in females. Despite these well-documented differences, most RCC clinical trials have not stratified outcomes by sex. The ARON-1 clinical study [27] directly addresses this gap by investigating sex-dependent treatment responses in advanced ccRCC, comparing ICI, TKI, and combination therapy across male and female cohorts, and provides the primary clinical benchmark against which the present model is calibrated.

1.3 Motivation for Computational Modeling

The complexity of the RCC tumor microenvironment—with heterogeneous cell populations, spatial organization, stochastic interactions, metabolic competition, and hormonal modulation—presents a challenge that no single experimental or analytical approach can fully address. Clinical trials are expensive, time-consuming, and limited in the treatment combinations and patient subpopulations they can explore. Computational modeling offers a complementary path by enabling *in silico* experimentation with

thousands of virtual patients, systematic exploration of inaccessible parameter spaces, algorithmic treatment optimization, and generation of mechanistic hypotheses for experimental testing. In RCC specifically, such models may address the biomarker gap by simulating how patient-specific variables interact to determine treatment response, identifying subpopulations for whom particular therapies are most likely to succeed.

Previous work by Merelli et al. [17] demonstrated the feasibility of agent-based RCC modeling using the Mesa framework, reproducing qualitative features of RCC progression on a two-dimensional lattice with simplified immune cell types. However, their model was limited to single-process execution, two-dimensional geometry, and lacked metabolic competition, sex hormone modulation, or clinical calibration. The present work extends that foundation in five directions: porting to Repast4Py for MPI-based parallelism, adding a continuous three-dimensional glucose field based on Fick's diffusion, introducing a 16-gene DNA system with codon-level mutation mechanics, incorporating sex hormone modulation of immune responses, and calibrating against ARON-1 clinical trial data using Bayesian optimization. Figure 1.2 provides a high-level overview of this modeling approach.

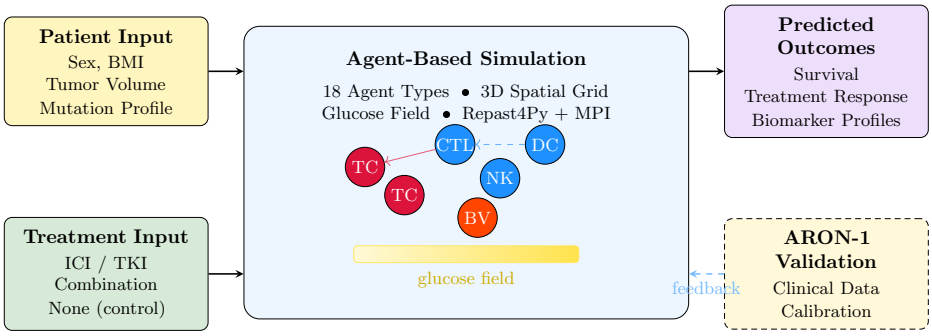

Figure 1.2: Conceptual overview of the RCC agent-based modeling approach.

1.4 Agent-Based Modeling Approach

Agent-based modeling (ABM) is a computational paradigm in which a system is represented as a collection of autonomous entities—agents—that interact with each other and their environment according to defined rules, with macroscopic behavior emerging from the aggregate of local interactions. ABM is particularly well suited to the tumor microenvironment because each agent can carry its own state—a unique genome, activation level, spatial position, and accumulated neighbor effects—providing natural representation of cellular heterogeneity. Agents occupy positions in a 3D grid and interact with their local neighborhood, capturing spatial phenomena such as immune-excluded regions, vascular networks, and glucose gradients. Biological processes like division, apoptosis, and immune recognition are inherently probabilistic, and ABM naturally incorporates this stochasticity at the individual level, allowing complex phenomena such as immune evasion and treatment resistance to emerge from simple local rules without explicit programming.

The choice of Repast4Py [8] as the simulation framework is motivated by its support for MPI-based parallel execution, enabling the model to scale to larger grid sizes and agent populations than single-process frameworks such as Mesa [14]. Repast4Py provides core abstractions—`SharedContext`, `SharedGrid`, and MPI-aware scheduling—

that distribute agents across processes while maintaining consistent spatial queries and inter-agent communication.

1.5 Project Objectives

This work pursues four principal objectives. First, the existing Mesa model is ported to Repast4Py for MPI parallelism, restructuring the agent hierarchy, spatial indexing, and scheduling while preserving all 18 agent types and their interaction logic. Second, a novel continuous 3D glucose concentration field is overlaid on the agent grid, where blood vessels inject glucose that diffuses via Fick's law and is consumed by all cell types at metabolically appropriate rates, introducing 8 new parameters governing glucose dynamics. Third, the model's predictions are validated against ARON-1 clinical trial data with particular attention to sex-stratified treatment responses. Fourth, an Optuna-based Bayesian optimization pipeline searches the parameter space for configurations reproducing clinical observations.

1.6 Report Structure

The report proceeds from theoretical background (Chapter 2) through model architecture and biological subsystems (Chapters 3–4), treatment mechanisms (Chapter 5), implementation details (Chapter 6), and parameter optimization (Chapter 7), before presenting simulation results and validation (Chapter 8) and concluding with future directions (Chapter 9).

2. Theoretical Background

> The cell is a machine driven by energy.
>
> — Albert Szent-Györgyi

THIS CHAPTER establishes the theoretical foundations of the RCC agent-based model, covering agent-based modeling and Repast4Py, the immunology and metabolism of the tumor microenvironment, the treatment modalities implemented in the simulation, and the related work that positions this thesis within the computational oncology literature.

2.1 Agent-Based Modeling

An agent-based model (ABM) consists of autonomous agents, a shared environment, and behavioral rules. System-level behavior is emergent rather than prescribed by global equations, making ABMs powerful for systems with heterogeneity, spatial structure, and nonlinear feedback—hallmarks of the tumor microenvironment. In contrast to ODE/PDE approaches that describe aggregate state variables, ABMs represent each entity individually (Figure 2.1). In this model, tumor cells carry individual 16-gene DNA sequences, agents must be in spatial proximity to interact, and mutation and immune recognition follow per-agent probabilistic rules. This richness comes at a computational cost scaling with population size, motivating MPI-based parallelization via Repast4Py.

Repast4Py Framework. Repast4Py [9] is a Python-based ABM toolkit that provides native MPI-based parallelism, distinguishing it from single-process frameworks such as Mesa or NetLogo. The simulation is distributed across MPI ranks, each responsible for a spatial partition and the agents within it. Agents are placed on a `SharedGrid`—a discrete N-dimensional lattice supporting multiple occupancy—and interact only with nearby agents, including read-only ghost copies of agents on neighboring ranks. This architecture enables near-linear scaling for spatially local interaction models like the tumor microenvironment. Chapter 6 provides a detailed treatment of the framework's internals and their application in this work.

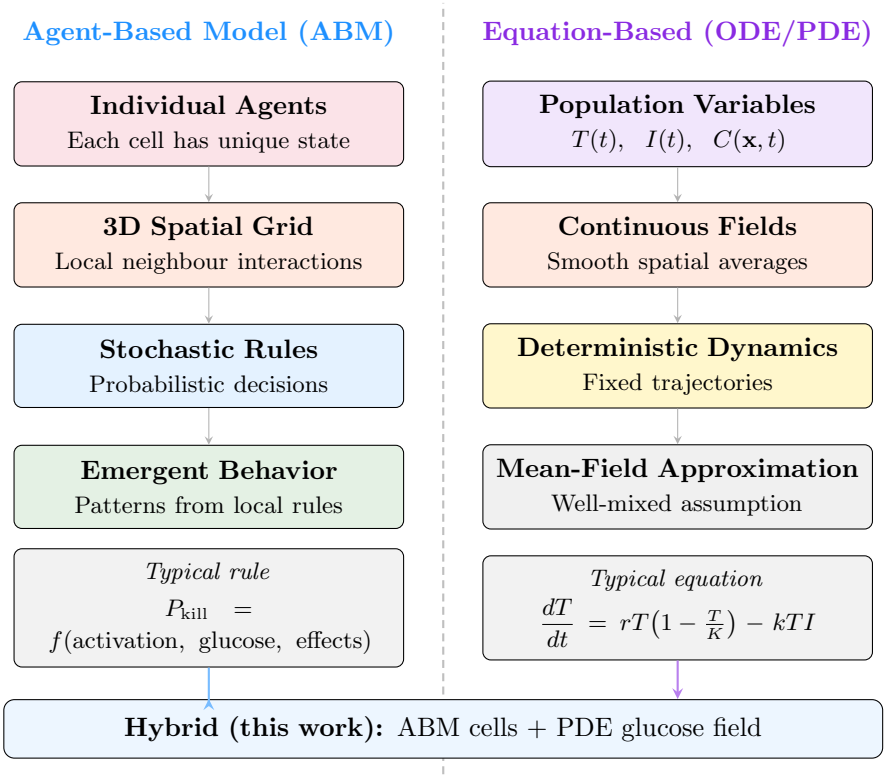

Figure 2.1: Comparison of agent-based and equation-based modeling paradigms. ABMs represent individual agents with local interaction rules, producing emergent system-level behavior; ODE/PDE models describe aggregate state variables through global equations.

2.2 Immunology of Renal Cell Carcinoma

Renal cell carcinoma is one of the most immunogenic solid tumors, a property rooted in the frequent inactivation of the von Hippel–Lindau (VHL) tumor suppressor gene (70–80% of clear cell cases) [11]. VHL loss stabilizes hypoxia-inducible factors, driving VEGF overexpression and hypervascularization that facilitates immune cell infiltration. The resulting paradox—abundant tumor-infiltrating lymphocytes coexisting with progressive tumor growth—reflects the interplay between immune activation and tumor-mediated evasion that the model seeks to capture. High mutational burden and a diverse neoantigen repertoire further make RCC amenable to checkpoint immunotherapy [19]. Table 2.1 summarizes the thirteen immune cell types represented in the model, spanning both innate and adaptive arms of the anti-tumor response.

Among these, the PD-1/PD-L1 checkpoint axis is most clinically relevant: PD-L1 on tumor cells engages PD-1 on T cells, suppressing proliferation and cytotoxic activity—a mechanism directly targeted by the ICI treatment modeled in this work. Figure 2.2 illustrates the temporal progression from innate recognition through adaptive activation to effector-mediated killing, including the immunosuppressive feedback loops that can blunt the response.

The cascade can equivalently be viewed as a sequential process with regulatory feedback (Figure 2.3), where each step depends on the preceding one, making the response vulnerable to disruption at multiple points.

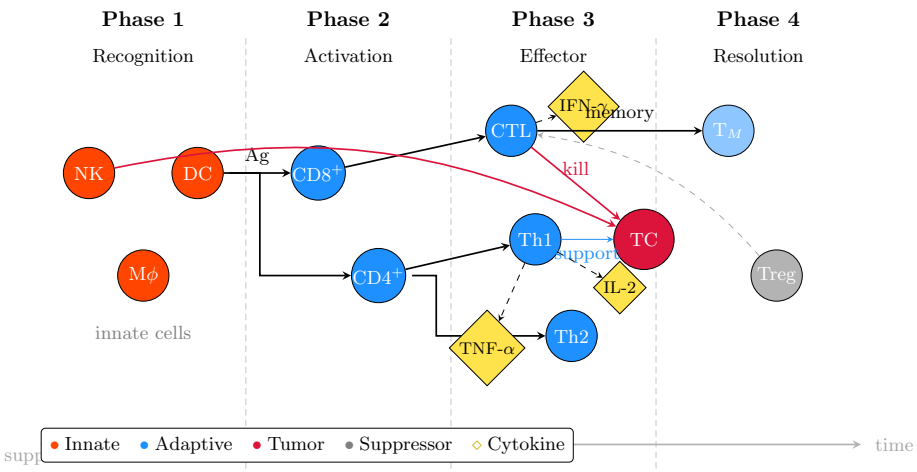

Figure 2.2: Immune response cascade from innate recognition through adaptive activation to effector-mediated tumor killing. Cytokines (IFN-γ, IL-2, TNF-α) mediate cross-talk between phases. Regulatory T cells and M2 macrophages provide immunosuppressive feedback that can blunt the effector response.

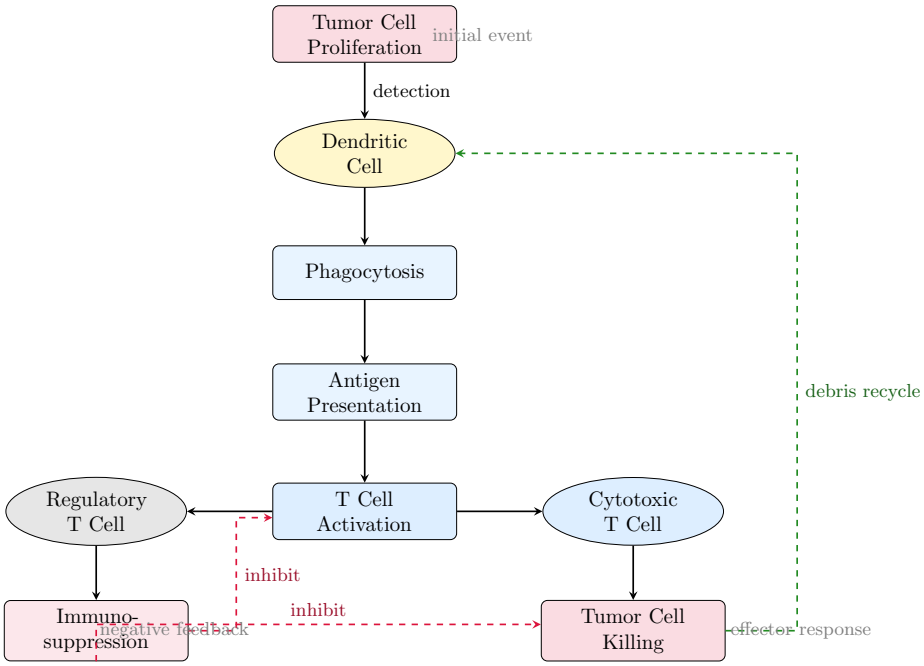

Figure 2.3: BPMN-style representation of the immune response cascade. Sequential processing from tumor detection through antigen presentation to T cell activation, with negative feedback from regulatory T cells (red dashed arrows) that can suppress effector functions at multiple points.

Table 2.1: Immune cell types and primary functions in the RCC model.

Cell Type	Primary Function
CD8+ Cytotoxic T Cell	Direct tumor killing
CD8+ / CD4+ Naive T Cell	CTL / helper precursors
Th1 / Th2 Helper T Cell	Pro-/anti-inflammatory
Regulatory T Cell	Immunosuppression
Dendritic Cell / pDC	Antigen presentation, T cell & NK activation
M1 / M2 Macrophage	Anti-tumor killing / pro-tumor promotion
Natural Killer Cell	Innate tumor killing
Mast Cell / Neutrophil	Context-dependent / sentinel

2.3 Tumor Metabolism

Cancer cells preferentially metabolize glucose through glycolysis even when oxygen is plentiful—the Warburg effect [15]. Although glycolysis yields only 2 ATP per glucose (vs. 36 from oxidative phosphorylation), it proceeds faster and feeds biosynthetic pathways for rapid division [29]. Activated T cells undergo similar metabolic reprogramming [5], creating competition for the same glucose supply: tumor growth depletes glucose, suppresses immune function, and permits further growth. This is especially relevant for immunotherapy, because checkpoint inhibitors can only restore T cell function if sufficient metabolic resources remain. The model captures this through an explicit glucose diffusion field (Chapter 4).

2.4 Treatment Modalities

The model implements two drug classes and their combination. Immune checkpoint inhibitors (ICIs) block the PD-1/PD-L1 axis, restoring T cell effector function against tumor cells [23, 24]. Tyrosine kinase inhibitors (TKIs) suppress VEGF-driven angiogenesis, restricting nutrient supply to the tumor while also inducing immunogenic cell death that primes dendritic cells for adaptive immune activation. Combination ICI+TKI therapy—now the standard first-line treatment for advanced RCC following pivotal trials such as KEYNOTE-426 [26] and JAVELIN Renal 101 [21]—exploits the synergy between these mechanisms: TKI-mediated vascular normalization improves immune cell access to the tumor, while ICI ensures that activated T cells retain cytotoxic capacity. The ARON-1 study [27] extends this line of investigation by examining whether the optimal strategy depends on patient sex. Chapter 5 provides the full mechanistic detail of how these treatments are realized in the simulation.

2.5 Related Work

This section positions the present work within the broader computational oncology landscape, covering continuum models, agent-based frameworks, and the direct predecessor of this thesis.

2.5.1 Continuum and PDE-Based Tumor Models

Classical mathematical oncology relies on ODE/PDE systems to describe tumor growth as a continuum process. Benzekry et al. [3] showed that standard growth laws (exponential, logistic, Gompertz) can reproduce aggregate growth curves but treat the tumor as a homogeneous mass, discarding intratumoral heterogeneity, spatial gradients, and stochastic cell-level events. Macklin et al. [16] advanced the paradigm by coupling reaction-diffusion nutrient transport with tissue mechanics, capturing macroscopic features like necrotic core formation, but their continuum representation still averages over individual cell states. These limitations motivate agent-based approaches whenever cell-level diversity—such as the coexistence of immunogenic and immune-evasive subclones, or stochastic T cell activation and exhaustion—is central to the question under study.

2.5.2 PhysiCell: A Continuous Mechanics-Based Cell Simulator

PhysiCell [10] represents cells as off-lattice agents governed by continuous mechanics (adhesion, repulsion, deformation) coupled with PDE-solved chemical fields. It excels at biophysical realism—spheroid compaction, migration through extracellular matrix—but its computational cost scales with mechanical solver fidelity, and its architecture is oriented toward biomechanical rather than immunological questions. The present work trades sub-cellular mechanical resolution for the ability to represent thirteen distinct immune cell types, gene-level tumor heterogeneity, and metabolic competition on a shared glucose field.

2.5.3 HAL: Hybrid Automata Library

The Hybrid Automata Library (HAL) [4] is a Java-based toolkit that combines on-lattice agents with continuous PDE layers for diffusible substances, sharing the conceptual approach of coupling discrete cells with a nutrient field. However, HAL models typically aggregate immune cells into a single generic effector population, whereas the present model distinguishes thirteen subtypes with specific behavioral rules. HAL also lacks native parallelism, limiting scalability for the large parameter sweeps required by Bayesian calibration.

2.5.4 Agent-Based Models of Cancer

Norton et al. [22] showed that spatial patterns of immune infiltration determine immunotherapy outcomes in breast cancer. Wang et al. [30] demonstrated cross-scale feedback by embedding intracellular signaling ODEs within agents, a philosophy adopted here through the 16-gene DNA system. Anderson and Quaranta [2] showed that spatial structure alters subclone competition. The present work builds on these foundations while targeting RCC with immune cell diversity exceeding most prior models.

2.5.5 The Merelli et al. Mesa-Based RCC Model

The most direct predecessor is the RCC ABM by Merelli et al. [17], implemented in Mesa on a 2D lattice with tumor cells and a simplified immune set. That model reproduced qualitative RCC progression features but lacked metabolic modeling, genetic

heterogeneity, and quantitative validation. The present thesis extends this in five directions: Repast4Py with MPI parallelism; a diffusion-based glucose field; a 16-gene DNA system; sex hormone modulation enabling ARON-1 investigation; and Bayesian optimization-based calibration against clinical endpoints.

2.5.6 Comparative Summary

Table 2.2 summarizes these trade-offs. The present work occupies a distinct position, prioritizing immunological detail, metabolic realism, and genetic heterogeneity within a scalable parallel framework.

Table 2.2: Comparison of the present model with related frameworks and the predecessor model.

Dimension	**This work**	**PhysiCell**	**HAL**	**Merelli et al.**
Agent types	18 (tumor, 13 immune, 4 environmental)	Custom cell phenotypes	On-/off-lattice agents	Tumor + few immune types
Spatial model	3D discrete lattice (Repast4Py grid)	Continuous off-lattice	Lattice + off-lattice	2D discrete lattice (Mesa)
Metabolic layer	Diffusion-based glucose field	PDE-based oxygen/nutrients	PDE-based substances	None
Immune detail	13 subtypes with specific rules	Generic effectors, customizable	Typically aggregated	Simplified immune types
Genetic model	16-gene binary DNA per tumor cell	Phenotype parameters	Not standard	None
Treatment modeling	ICI + TKI + hormone modulation	Drug diffusion, pharmacokinetics	Drug fields	Basic treatment rules
Parallelism	MPI-native	OpenMP threads	Single-process	Single-process
Validation approach	Bayesian opt. + ARON-1 clinical data	Problem-specific	Problem-specific	Qualitative plausibility

No prior ABM of the tumor microenvironment combines explicit glucose metabolism, sex hormone modulation, and calibration against sex-stratified clinical data within a single framework. The present work takes an early step toward precision oncology by calibrating against sex-stratified ARON-1 data, demonstrating that even a single stratification variable yields clinically meaningful differences in predicted treatment response.

With these theoretical foundations in place, the following chapter presents the overall model architecture and design decisions that translate these concepts into a working simulation.

3. Model Description

> The purpose of abstraction is not to be vague, but to create a new semantic level in which one can be absolutely precise.
>
> — Edsger W. Dijkstra

THIS CHAPTER presents the architecture of the agent-based model from the ground up: the spatial environment in which agents operate, the class hierarchy that organizes 18 agent types into a coherent inheritance tree, the paracrine effect system that mediates cell–cell signaling, the movement mechanics that govern spatial behavior, the step-by-step execution loop that advances the simulation, and the terminal conditions that determine patient outcomes.

3.1 Spatial Environment

The simulation takes place on a three-dimensional discrete grid whose side length is derived from the patient's tumor volume. Given a volume V in cm^3 and a block size $\Delta = 10\ \mu\text{m}$, the grid dimension is

$$L = \left\lfloor \frac{(V \times 10^{12})^{1/3}}{\Delta} \right\rfloor \tag{3.1}$$

producing a cubic $L \times L \times L$ lattice (e.g., $L = 100$ for $V = 0.001\ \text{cm}^3$). Multiple agents may occupy the same voxel; boundaries are *sticky*. A continuous glucose concentration field $C(\mathbf{x}, t) \in \mathbb{R}_{\geq 0}$ overlays the grid and evolves according to a reaction–diffusion equation (Chapter 4), providing the metabolic context for proliferation, killing, and chemotactic navigation.

3.2 Agent Architecture

3.2.1 Class Hierarchy

The 18 agent types are organized into a four-level inheritance tree rooted at Repast4Py's `Agent` class (Figure 3.1). Two abstract base classes introduce the shared functionality that makes the hierarchy tractable. **Cell** provides position tracking, deferred movement, effect collection, neighborhood queries, and duplication logic for every spatially embodied agent. **TCell** extends `Cell` with T lymphocyte-specific behaviors: a 16-bit TCR,

antigen matching via Hamming distance ($d_H \leq 3$), hormone perception with exponential decay, and 17 sex-hormone modulation channels. Five T cell types inherit from `TCell`; eleven cellular agents inherit directly from `Cell`; and two non-cellular agents (`SexHormone`, `Cytokine`) inherit from the framework's `Agent` without spatial semantics. Three cellular agents (`DendriticCell`, `PlasmacitoidDendriticCell`, `MacrophageM1`) additionally incorporate a `PhagocyticMixin` for antigen processing.

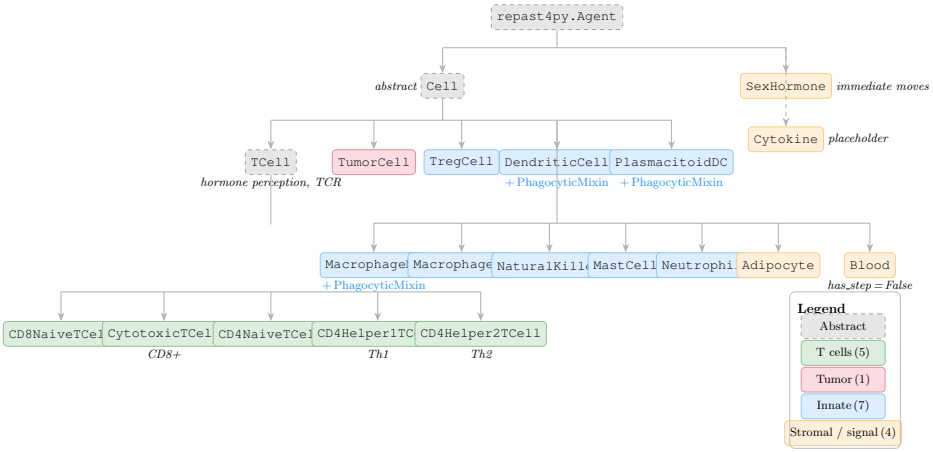

Figure 3.1: Agent class hierarchy. Abstract bases are shown with dashed borders. Color encodes functional category: T cells (green), tumor (red), innate immune (blue), and stromal/signaling (orange). Agents marked *+ PhagocyticMixin* gain a hunt–engulf–present state machine.

3.2.2 Agent Types

Table 3.1 lists all 18 agent types with their enumeration IDs and functional categories. Each type is identified at runtime by an integer from the `AgentType` enumeration, enabling O(1) type checks during neighborhood queries and effect collection.

Table 3.1: Agent types in the RCC model, with enumeration IDs and functional categories.

ID	Agent Type	Class Name	Category
0	Tumor Cell	`TumorCell`	Neoplastic
1	Cytotoxic T Cell	`CytotoxicTCell`	Adaptive
2	$CD8^+$ Naïve T Cell	`CD8NaiveTCell`	Adaptive
3	$CD4^+$ Naïve T Cell	`CD4NaiveTCell`	Adaptive
4	$CD4^+$ Th1 Cell	`CD4Helper1TCell`	Adaptive
5	$CD4^+$ Th2 Cell	`CD4Helper2TCell`	Adaptive
6	Regulatory T Cell	`TregCell`	Adaptive
7	Dendritic Cell	`DendriticCell`	Innate
8	Plasmacytoid DC	`PlasmacitoidDendriticCell`	Innate
9	M1 Macrophage	`MacrophageM1`	Innate
10	M2 Macrophage	`MacrophageM2`	Innate
11	Natural Killer	`NaturalKiller`	Innate
12	Mast Cell	`MastCell`	Innate
13	Neutrophil	`Neutrophil`	Innate
14	Adipocyte	`Adipocyte`	Stromal
15	Blood Vessel	`Blood`	Vasculature
16	Sex Hormone	`SexHormone`	Signaling
17	Cytokine	`Cytokine`	Signaling

3.2.3 Behavioral Flags

Three boolean class-level flags govern loop participation: `has_effect` (10 types emit paracrine effects), `needs_effect` (8 types collect neighbor effects), and `has_step` (all except `Blood`, which is static). These flags let the model skip unnecessary work per tick.

3.3 Agent Interactions

The 18 agent types interact through four principal mechanisms (Figure 3.2). *Cytotoxic killing*: CTLs, NK cells, M1 macrophages, and dendritic cells can directly destroy tumor cells, with kill probability modulated by the local effect field, glucose availability, and—for CTLs—TCR–neoantigen affinity (Hamming distance). *Activation and differentiation*: dendritic cells activate naïve $CD8^+$ and $CD4^+$ T cells upon phagocytosis; activated $CD8^+$ cells become CTLs, while $CD4^+$ cells differentiate into Th1, Th2, or Treg depending on the local cytokine milieu. *Immunosuppression*: Tregs suppress CTL killing (-0.3 on `t_kill_rate_effect`) and DC phagocytosis; M2 macrophages shield tumor cells and promote angiogenesis; tumor cells themselves inhibit CTLs via PD-L1 (CD274-dependent). *Promotion and recruitment*: Th1 cells recruit M1 macrophages and DCs; pDCs spawn NK cells; M2 macrophages promote angiogenesis; adipocytes drive M2 polarization.

Table 3.2 summarizes the key interactions by agent type.

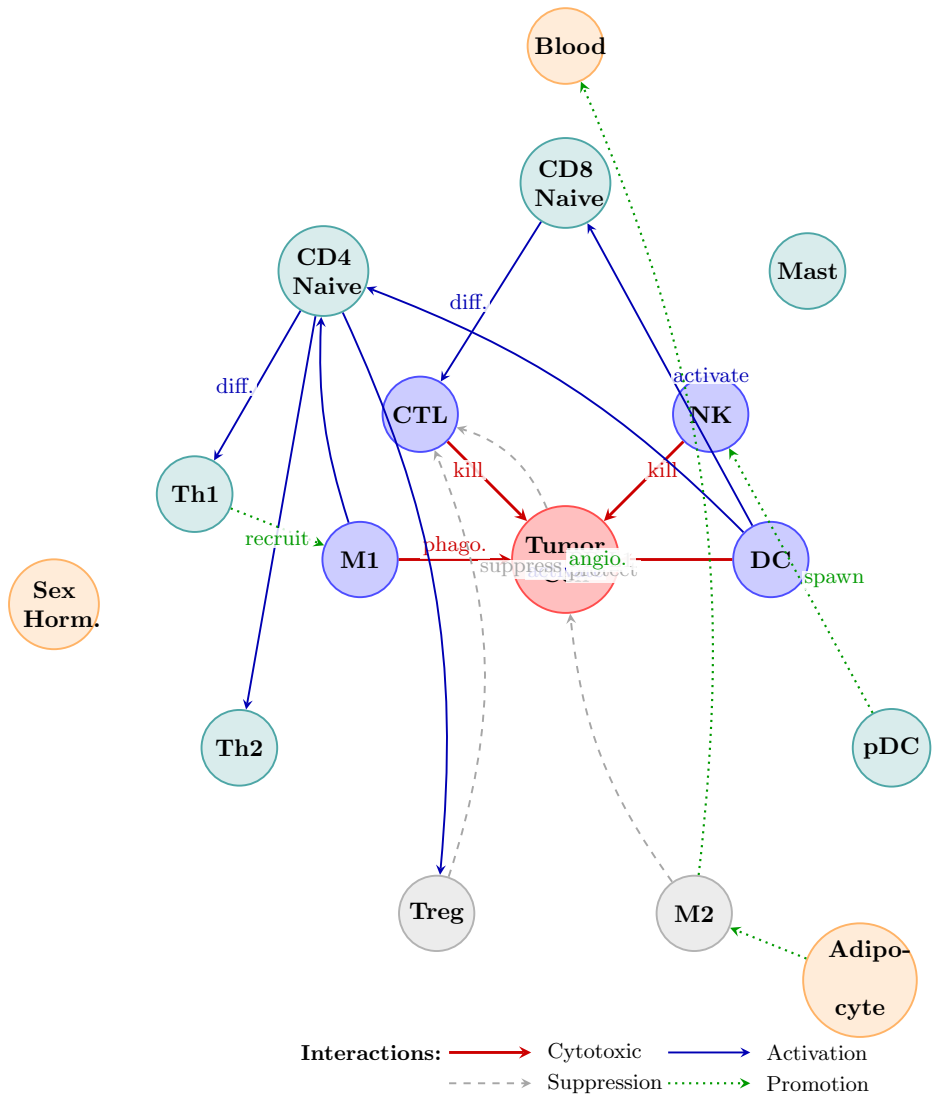

Figure 3.2: Hub-and-spoke interaction network. The tumor cell sits at the center; the inner ring contains the four direct killer types; the outer ring holds support and regulatory cells. Edge color encodes interaction type: cytotoxic (red), activation (blue), suppression (grey dashed), and promotion (green dotted).

Table 3.2: Summary of agent interactions in the RCC model.

Agent	**Role**	**Key Effects**
TumorCell	Neoplastic	DNA-driven growth; PD-L1 evasion; 200× glucose uptake
CTL	Cytotoxic	TCR-matched killing; glucose/hormone-modulated
CD8/CD4 Naïve	Precursor	Differentiate upon DC activation
Th1	Pro-inflam.	Recruits M1, DC; boosts tumor apoptosis
Th2	Mixed	Mild growth boost; recruits CTLs
Treg	Suppressive	Inhibits CTL (−0.3), proliferation, DC phagocytosis
DC	APC	Phagocytoses tumor; activates naïve T cells
pDC	Specialized	Spawns NK; boosts killing; promotes Treg diff.
M1	Anti-tumor	Phagocytoses tumor; boosts Th1; can polarize to M2
M2	Pro-tumor	Inhibits killing; promotes growth & angiogenesis
NK	Innate cyto.	Kills tumor; glucose-dependent
MastCell	Dual role	50/50 pro-/anti-tumor phenotype per step
Neutrophil	Sentinel	Age-limited lifespan
Adipocyte	Stromal	Mild tumor promotion; M2 polarization
Blood	Vasculature	Glucose source; static; angiogenesis target
SexHormone	Signaling	17 immune modulation channels
Cytokine	Signaling	Reserved for future extension

3.4 Effect System

The paracrine effect system is the primary mechanism through which agents influence one another without direct contact. Each `Cell` agent maintains an `experienced_effects` object containing 14 floating-point fields (Figure 3.3), grouped into five functional categories: T cell modulation (4 fields controlling kill rate, proliferation, apoptosis, and activation), helper-specific effects (2 fields for Th1/Th2 proliferation), regulatory effects (3 fields governing Treg differentiation and M1/M2 macrophage polarization), tumor and vasculature effects (3 fields for tumor growth, apoptosis, and angiogenesis), and specialized effects (2 fields for DC phagocytosis and NK kill rate).

3.4.1 Effect Production and Collection

At each step, the model iterates over agents with `needs_effect` set, querying the Moore neighborhood at radius 1, retrieving cached effect objects, and aggregating via element-wise addition. The aggregation is additive: a Treg (−0.3 on kill rate) and an M1 macrophage (+0.1) near a CTL produce a net −0.2. This superposition means no agent requires global knowledge; phenomena such as immune exclusion zones emerge from the spatial arrangement of producers.

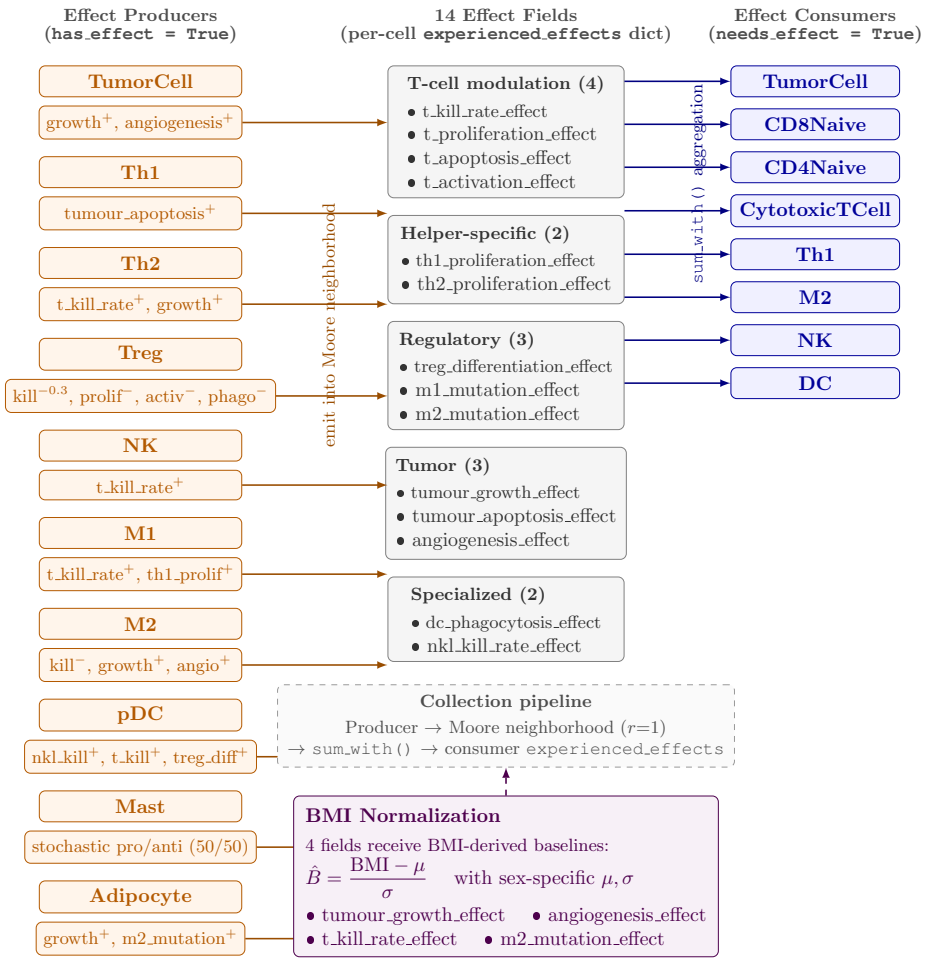

Figure 3.3: The paracrine effect system. Ten producer types (left) emit effects into the Moore neighborhood. The 14 effect fields (center) accumulate via element-wise addition. Eight consumer types (right) read the aggregated effects to modulate their behavior. BMI normalization (bottom) provides baseline offsets for four fields.

3.4.2 BMI Normalization

Four of the 14 effect fields receive BMI-dependent baseline offsets. The normalized BMI $\hat{B} = (\text{BMI} - \mu)/\sigma$ uses sex-specific parameters ($\mu = 26.5$, $\sigma = 5.5$ for females; $\mu = 27.5$, $\sigma = 4.5$ for males). The offset is applied as $\pm\hat{B} \times w_i$ to Treg differentiation ($-$), M1 polarization ($-$), M2 polarization ($+$), and NK kill rate ($-$). A positive $\hat{B}$ (above-average BMI) therefore shifts the microenvironment toward immunosuppression—more M2, more Treg, less NK killing—consistent with clinical evidence linking obesity to poorer RCC prognosis.

3.5 Movement Mechanics

Spatial behavior in the model follows a two-phase *deferred movement* pattern: during the agent-stepping phase, each agent computes its desired next position and stores it in `_desired_pos`; after all agents have stepped, the model batch-processes all pending

moves in a single pass. This ensures that every agent makes its movement decision based on the same spatial snapshot, eliminating bias from the random activation order.

3.5.1 Directed Movement

The primary navigation method, `move_towards()`, uses an expanding-radius search: it queries the spatial index at radii 1, 2, 4, 8, ... (doubling each iteration), finds the closest target by Manhattan distance $d_M(\mathbf{a}, \mathbf{b}) = |a_x - b_x| + |a_y - b_y| + |a_z - b_z|$, computes the normalized Euclidean direction vector, and moves one step along it (clamped to grid boundaries). The doubling strategy guarantees $O(\log L)$ worst-case iterations. The maximum search radius scales dynamically with tumor population as $\max(5,\ N_{\text{tumor}}/(10 \cdot w_{\text{search_dim}}))$, allowing immune cells to detect growing tumor masses at increasing distances.

3.5.2 Glucose Chemotaxis

When an immune cell (CTL, M1, DC, or NK) fails to find a direct target, it may fall back to glucose chemotaxis. With probability $w_{\text{glucose_chemotaxis_strength}}$, the agent queries the glucose field for the discretized gradient at its position:

$$(\Delta x, \Delta y, \Delta z) = \operatorname{sign}\big(\nabla C(\mathbf{x})\big) \tag{3.2}$$

where the gradient is computed via central finite differences with Neumann (zero-flux) boundary conditions. The agent then moves one step along the gradient direction, migrating toward glucose-rich regions—typically areas near blood vessels and active tumor growth fronts—providing a biologically plausible navigation strategy in the absence of direct antigen detection.

3.5.3 Sex Hormone Diffusion

Sex hormone agents follow a *non-deferred* movement model: each step, a hormone randomly selects (50/50) between biased diffusion away from vasculature (drift strength 0.25, weighting 26 Moore neighbors by Manhattan distance to the nearest blood vessel) and simple diffusion biased toward $+y/+z$ that removes the hormone upon grid exit. Moves execute immediately since hormones do not depend on each other's positions.

Figure 3.4 summarizes the complete movement decision process as a flowchart.

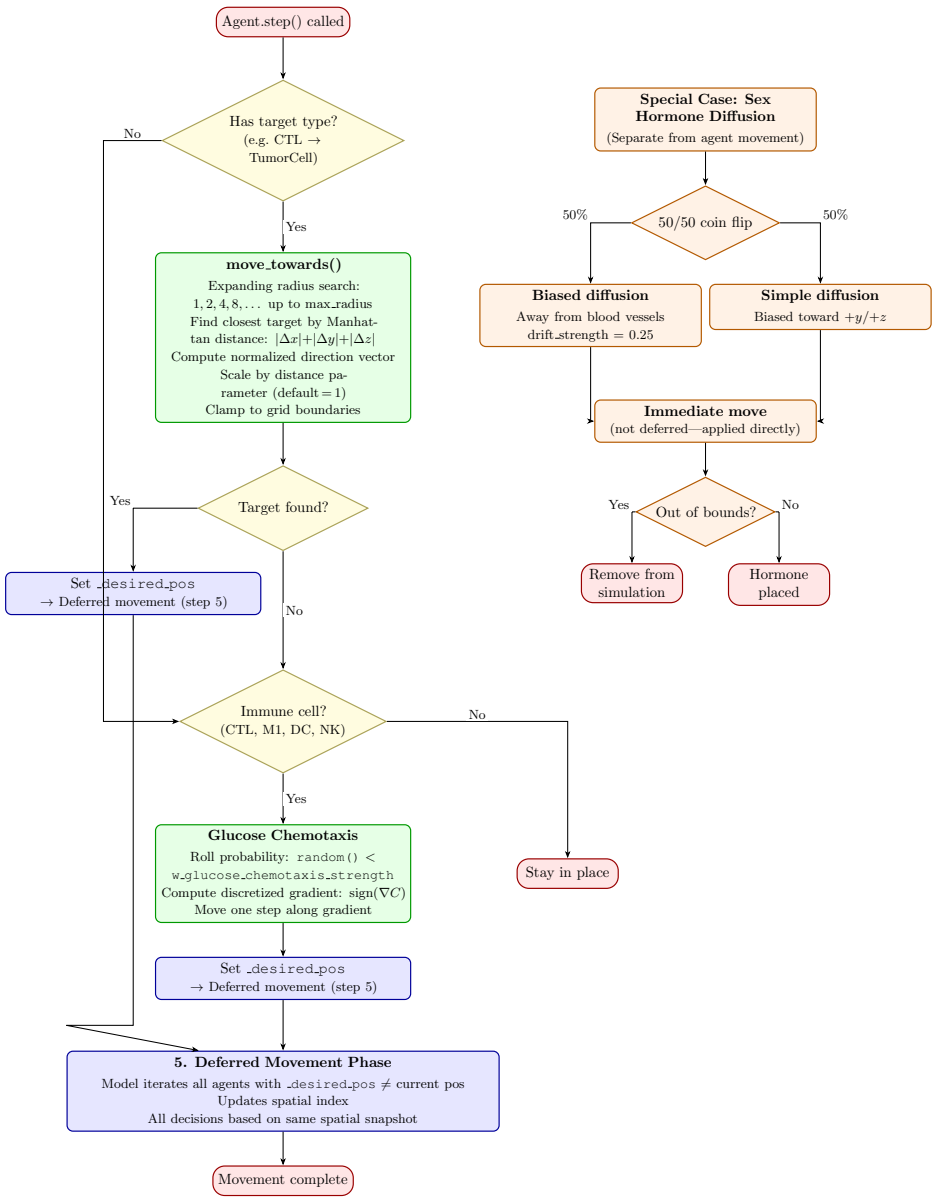

Figure 3.4: Movement decision flowchart. Cellular agents (left branch) attempt directed movement toward a target type, falling back to glucose chemotaxis if no target is found. All cellular moves are deferred. Sex hormones (right branch) use a biased random walk with immediate execution.

3.6 Simulation Step

Each call to `RCCModel.step()` advances the simulation by one discrete time unit through 13 operations executed in a fixed sequence (Figure 3.5). The ordering reflects three design constraints: (i) effects must be collected before agents act on them, (ii) treatment must be applied at the correct simulation time, and (iii) spatial updates must be deferred to prevent order-dependent artifacts.

1. **Collect data.** Record population counts by agent type, glucose field statistics (mean, min, max), and cumulative kill tallies. These are written to the data log at the end of each step.

2. **Check termination.** Evaluate the four terminal conditions (Section 3.7). If any is met, write the final log entry and halt the simulation.

3. **Reset immune infiltration.** Set the immune infiltration factor to 1 for Th1, Th2, and MastCell types, ensuring that these agents begin each step with a clean baseline before effects accumulate.

4. **Apply effects.** For each agent with `needs_effect = True`, collect `Effect` objects from the Moore neighborhood (radius 1) and aggregate them into the agent's `experienced_effects` via element-wise addition.

5. **Treatment step.** If the current step $t \geq t_{\text{start}}$, apply all active drug effects (ICI, TKI, or combination). Drug mechanisms are detailed in Chapter 5.

6. **Sex drift.** Apply sex-dependent modulation to the $CD8^+$ naïve T cell population: duplication for male patients, removal for female patients, with the magnitude decremented each step.

7. **Glucose field step.** Blood vessels inject glucose at cached positions; the field is diffused via a 3D discrete Laplacian (6-connected stencil, Neumann boundaries); natural decay reduces concentrations by factor $(1 - \lambda)$.

8. **Manage blood.** Evaluate angiogenesis conditions and spawn new blood vessels if warranted (minimum separation 30 cells, effect radius 3). Refresh tumor cell blood access via breadth-first search from vessel-connected sources.

9. **Update search dimension.** Recompute the global cell search dimension as $\max(5,\ N_{\text{tumor}}/(10 \cdot w_{\text{search_dim}}))$, ensuring that immune cells can detect tumors at ranges proportional to tumor burden.

10. **Shuffle and step agents.** Take a snapshot of all living agents with `has_step = True`, shuffle randomly, and invoke each agent's `step()` method. Agents set `_desired_pos` during this phase but are not yet moved.

11. **Deferred movement.** Batch-move all agents in the `_pending_moves` set whose `_desired_pos` differs from their current position, updating the spatial index in a single pass.

12. **Spawn hormones.** Generate sex hormone agents at blood vessel locations according to sex-specific hormone production probabilities.

13. **Increment step counter.** Advance $t \leftarrow t + 1$ and loop back to step 1.

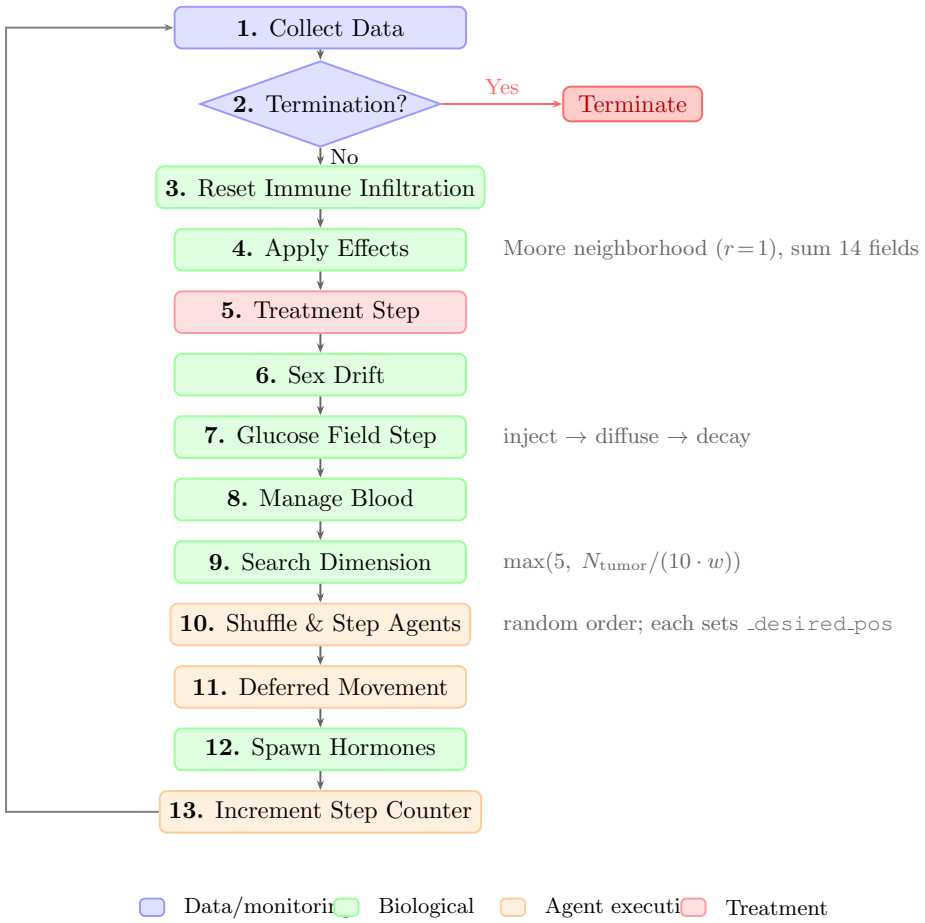

Figure 3.5: The 13-step simulation loop. Color encodes functional category: data and monitoring (blue), biological processes (green), agent execution (orange), and treatment (red). Step 2 is a decision point that terminates the simulation if any terminal condition is satisfied.

3.7 Terminal Conditions

The simulation terminates when any of four conditions is met: maximum steps reached ($t \geq t_{\max}$, death), tumor elimination (count $= 0$, survival), critical mass (≥ 2000 cells, death), or uncontrolled growth ($\bar{G} \geq w_{\text{growth_threshold}}$, death). The growth metric is

$$\bar{G} = \frac{1}{N_T} \sum_{i=1}^{N_T} \sigma\big(g_i + a_i;\ k{=}10\big) \tag{3.3}$$

combines effect-driven growth, DNA-derived proliferation, blood access, and apoptosis components. This last condition detects rapid acceleration even when absolute cell counts remain below critical mass.

With the model's spatial, behavioral, and interaction architecture established, the next chapter examines the three biological subsystems—glucose metabolism, DNA mutation, and sex hormones—that drive the emergent dynamics within this framework.

4. Biological Subsystems

> Nothing in biology makes sense
> except in the light of evolution.
>
> — Theodosius Dobzhansky

THE TUMOR MICROENVIRONMENT is governed by three tightly coupled biological subsystems that together determine the trajectory of disease progression and treatment response. First, glucose metabolism creates spatial resource gradients through the Warburg effect, establishing zones of nutrient abundance near vasculature and depletion around tumor masses; these gradients shape both tumor proliferation capacity and immune effector function. Second, a 16-gene DNA mutation system drives clonal evolution, enabling tumor cells to accumulate mutations that enhance proliferation, suppress apoptosis, and critically, evade immune recognition through loss of antigen presentation machinery. Third, circulating sex hormones—estrogen, testosterone, and progesterone—modulate the strength and character of the immune response through 17 distinct effect channels, producing the pronounced sex disparities observed clinically in renal cell carcinoma. These subsystems interact bidirectionally: glucose gradients impose the selective pressure under which genetic evolution proceeds, the resulting mutations determine each tumor cell's visibility to immune surveillance, and the hormonal milieu sets the immunological context within which recognition and killing occur. The following sections present each subsystem's biological basis, mathematical formulation, and computational implementation before examining how they couple to produce emergent dynamics.

4.1 Glucose Microenvironment

4.1.1 Biological Motivation

The metabolic landscape of the tumor microenvironment plays a decisive role in determining immune-tumor interactions. As depicted in Figure 4.1, Otto Warburg observed that tumor cells preferentially metabolize glucose through aerobic glycolysis, resulting in substantially elevated glucose uptake (up to 200-fold compared to normal tissue). This metabolic competition between tumor cells and infiltrating immune cells creates nutrient gradients that shape spatial organization, as regions surrounding tumor masses become nutrient-deprived. T cells and NK cells critically depend on glucose availability for cytotoxic activity — when glucose is scarce, T cell proliferation is impaired, IFN-γ production decreases, and cytotoxic granule release is diminished. In the absence of direct cell-cell contact, immune cells can follow glucose concentration gradients as a

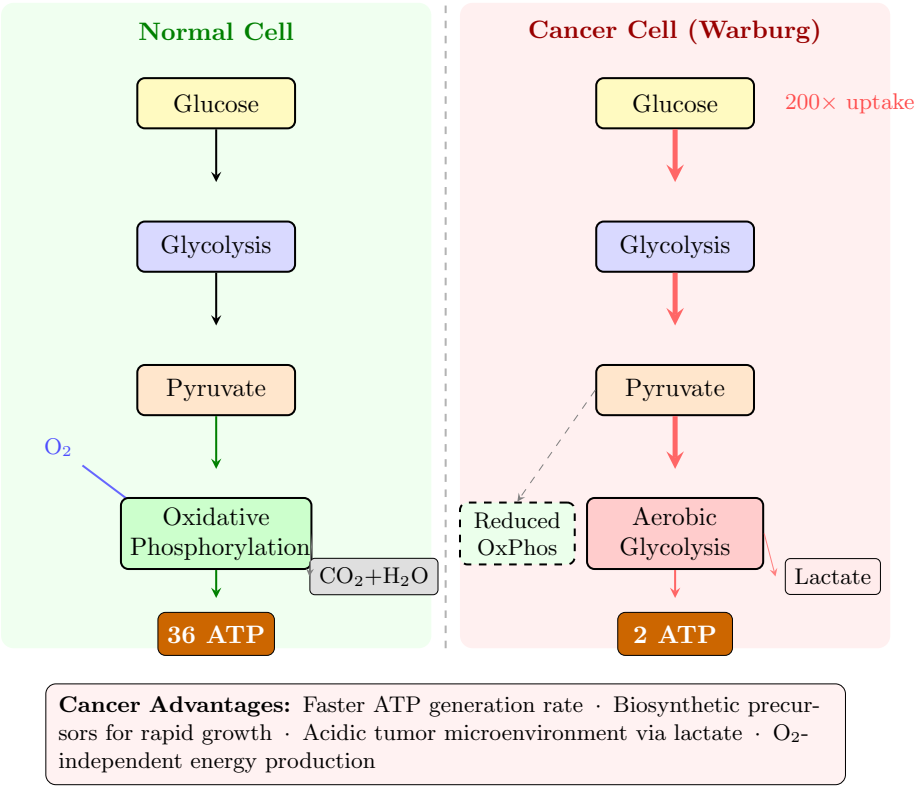

Figure 4.1: The Warburg effect: metabolic competition in the tumor microenvironment.

proxy for proximity to functional vasculature and viable tissue, providing a chemotactic fallback behavior.

The spatial distribution evolves dynamically as the simulation progresses (Figure 4.2).

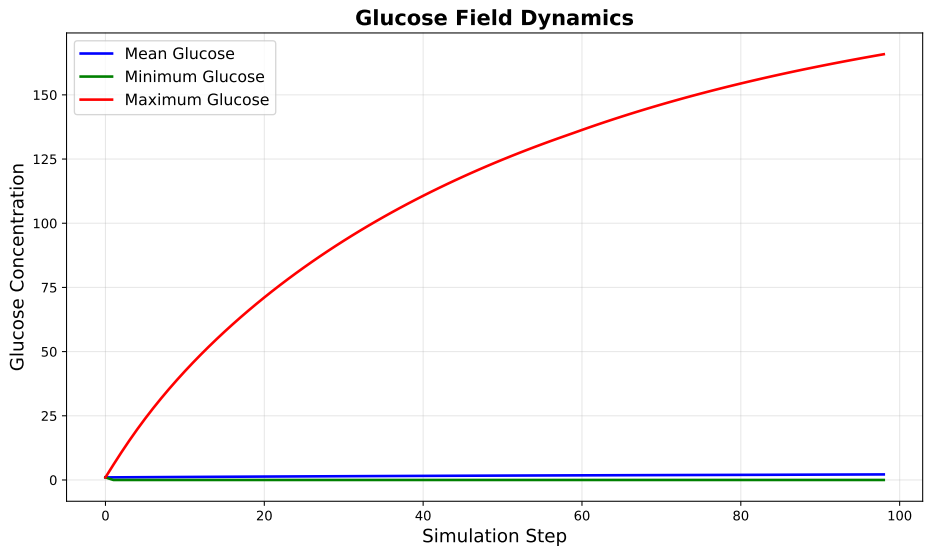

Figure 4.2: Glucose field dynamics showing depletion zones around tumor masses and gradients from blood vessels.

Blood vessels serve as point sources, the field diffuses according to Fick's law, and cellular agents consume glucose at position-dependent rates (Figure 4.3).

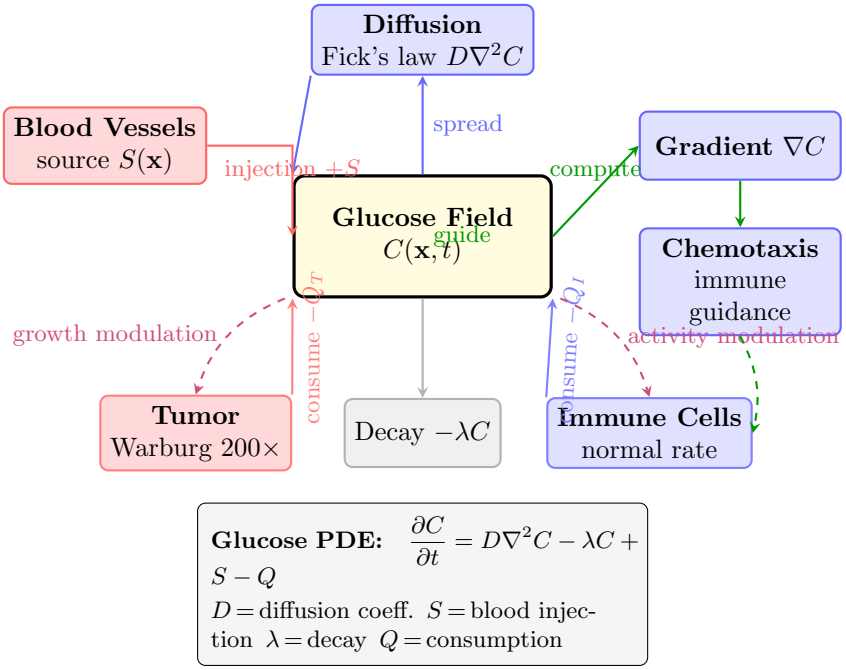

Figure 4.3: Glucose system architecture showing the diffusion field, sources from blood vessels, and consumption by cellular agents.

4.1.2 Mathematical Formulation

The glucose concentration field $C(\mathbf{x}, t)$ evolves according to a reaction–diffusion equation derived from Fick's second law:

$$\frac{\partial C}{\partial t} = D\,\nabla^2 C \;-\; \lambda\, C \;+\; S(\mathbf{x}) \;-\; Q(\mathbf{x}), \tag{4.1}$$

where $C(\mathbf{x}, t)$ is the glucose concentration, D is the diffusion coefficient (`w_glucose_diffusion`, default 0.1), λ is the natural decay rate (`w_glucose_decay`, default 0.01), $S(\mathbf{x})$ is the source term (glucose injection from blood vessels), and $Q(\mathbf{x})$ is the consumption term (uptake by tumor and immune cells).

The continuous Laplacian $\nabla^2 C$ is discretized on a regular 3D Cartesian grid using the standard 6-connected stencil, implemented as a 3D convolution using `scipy.ndimage.convolve`. The diffusion equation is advanced using explicit Euler integration with stability condition $D < 1/6 \approx 0.167$; the default $D = 0.1$ remains safely within the stable regime. Neumann (zero-flux) boundary conditions are enforced, corresponding to no glucose flux through simulation domain walls.

Source and Consumption Models. Blood vessel agents inject glucose at their grid positions at rate r_{source} (`w_glucose_source_rate`, default 5.0) per step. Tumor cells consume glucose reflecting the Warburg effect at rate r_{tumor} (`w_glucose_tumor_consumption`, default 2.0), while immune cells consume at lower rate r_{immune} (`w_glucose_immune_consumption`, default 0.5).

Tumor cell proliferation probability is modulated by glucose availability: $f_{\text{glucose}} = \min(1.0, C(\mathbf{x}_t) \cdot w_{\text{growth_sensitivity}})$, and immune cell kill rates are similarly scaled: $f_{\text{immune}} = \min(1.0, C(\mathbf{x}_i) \cdot w_{\text{immune_sensitivity}})$. When glucose is abundant ($C \geq 1.0$), activities proceed at full rate; as glucose drops below 1.0, functions are proportionally suppressed.

4.1.3 Gradient and Chemotaxis

The glucose gradient is computed using central finite differences, and for agent movement on the integer grid, the continuous gradient is converted to discrete step directions using the sign function. Immune cells that implement glucose chemotaxis (`CytotoxicTCell`, `MacrophageM1`, `DendriticCell`, `NaturalKiller`) employ a fallback strategy where they first attempt to move towards nearby tumor cells, and if no target is found, they move in the direction of increasing glucose concentration with probability p_{chemo} (`w_glucose_chemotaxis_strength`, default 0.3).

4.1.4 Implementation

The glucose system is encapsulated in the `GlucoseField` class, which stores concentrations as a 3D NumPy array (`float64`, initialized to 1.0). Each simulation step proceeds through injection (blood vessels deposit glucose at their grid positions), diffusion (discrete Laplacian convolution with Euler integration), and decay (multiplication by $1 - \lambda$). Agent-level consumption occurs within each agent's `step()` method via vectorized NumPy operations, and the class exposes aggregate statistics (mean, min, max) for tracking metabolic dynamics.

Mean glucose levels decline as tumor mass grows; minimum concentrations approach zero in regions of highest tumor density (Figure 4.4).

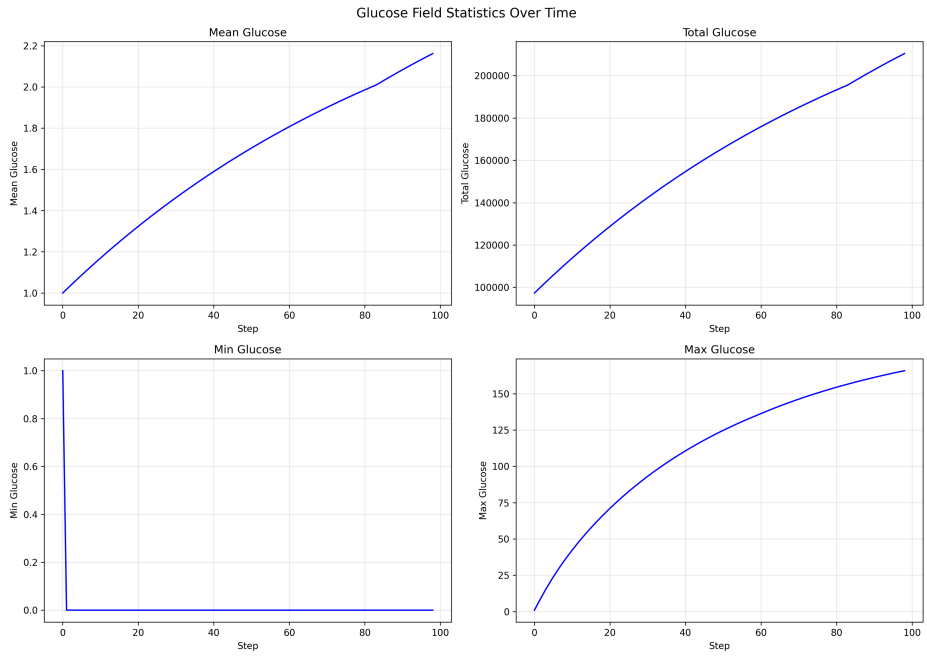

Figure 4.4: Glucose concentration time series (mean, min, max) reflecting injection, consumption, and diffusion.

The 3D field structure (Figure 4.5) produces clearly delineated metabolic territories that constrain agent behavior.

The circular coupling between injection, diffusion, consumption, and chemotaxis is summarized in Figure 4.6.

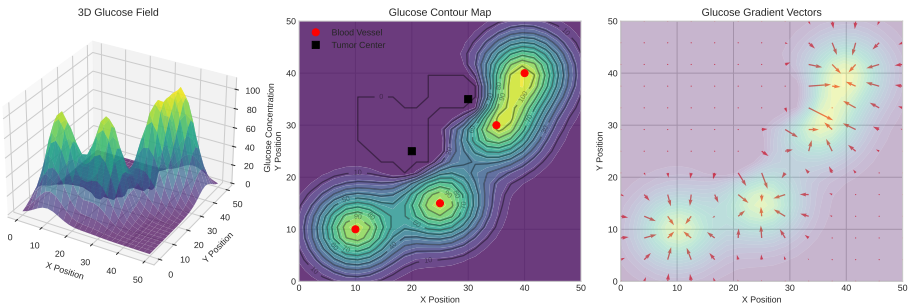

Figure 4.5: 3D glucose field: high concentrations (yellow) near blood vessels, depletion (blue) near tumor masses.

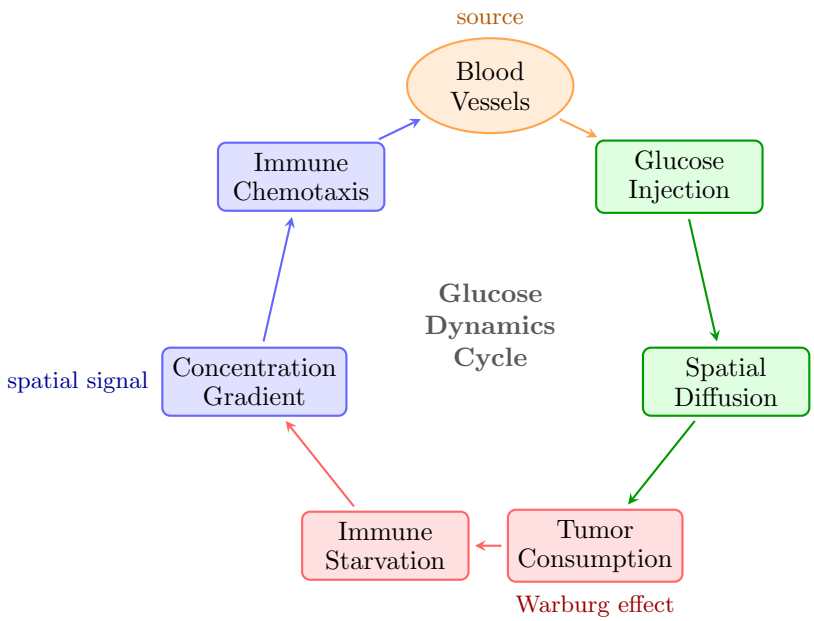

Figure 4.6: Glucose system cycle illustrating the circular flow of glucose dynamics in the tumor microenvironment.

4.2 DNA and Mutation System

The gene-level representation used by tumor cells encodes 16 cancer-relevant genes, supports codon-level translation, introduces mutations during cell division with genomic-instability-dependent rates, and derives phenotypic parameters that govern tumor behavior and immune evasion.

4.2.1 Gene Model

Each tumor cell carries a DNA object encoding 16 genes chosen for their relevance to renal cell carcinoma immunology, spanning antigen presentation, immune checkpoint signaling, DNA repair, tumor suppression, and oncogenic proliferation.

The 16 genes span five functional categories: *antigen presentation* (MHC, B2M, TAP1, TAP2, JAK1, JAK2)—mutations in these genes reduce surface peptide display and enable immune evasion; *immune checkpoint* (CD274/PD-L1)—overexpression engages PD-1 on T cells; *DNA repair* (BRCA1)—dysfunction increases genomic instability

and mutation rate; *tumor suppression* (TP53, RB1, CDKN2A, APC)—loss removes cell cycle checkpoints and apoptosis triggers; and *oncogenes* (MYC, RAS, EGFR, HER2)—gain-of-function drives proliferation.

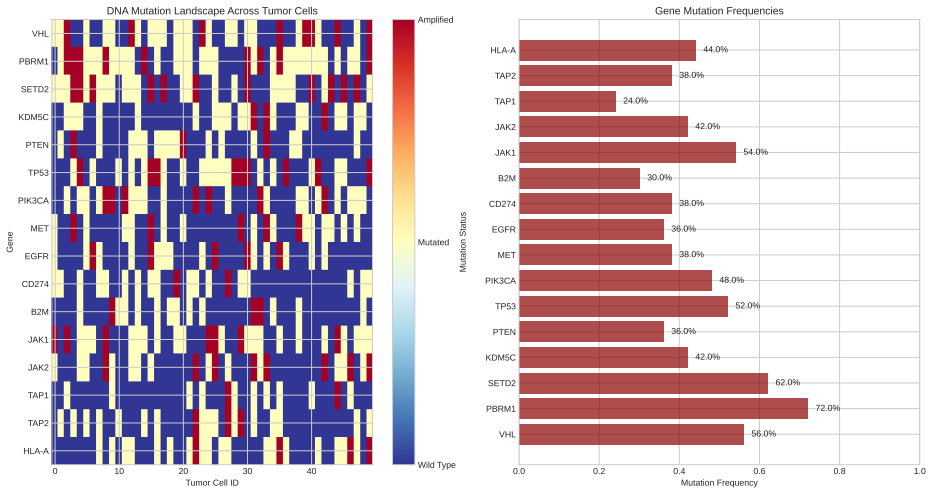

Figure 4.7: DNA mutation heatmap across the 16 genes; color intensity indicates mutation severity.

Each gene occupies a defined region within a synthetic DNA sequence assembled by concatenating spacer, promoter (including TATA box), coding region with codon-encoded wildtype protein, and stop codon, using full 64-codon translation.

4.2.2 Mutation Mechanics

Mutations accumulate through injected mutations at initialization and stochastic point mutations during DNA duplication. Genomic instability is controlled by BRCA1 mutation status, with mutation rate scaling as $\mu = (s_{BRCA1})^{\max(1, e_{BRCA1})} \times 0.05$. Initial mutations are sex-dependent: males start with 3 BRCA1 mutations, females with 1. This asymmetry compounds over time—higher initial BRCA1 dysfunction increases the per-nucleotide mutation rate during subsequent divisions, producing greater clonal diversity and faster immune evasion in male tumors.

4.2.3 Gene Expression and Phenotypic Effects

The lifecycle of a tumor cell—from quiescence through proliferation, immune interaction, and potential apoptosis—is governed by a state machine shown in Figure 4.8. Each gene's phenotypic contribution is determined by promoter expression (TATA motif count) and mutation score (Levenshtein distance from wildtype protein). The DNA system computes five aggregate phenotypic parameters:

Antigen Presentation Chance: $p_{AP} = \max(0, 6 - \frac{\sum_{g \in \{MHC, JAK1, JAK2, B2M, TAP1, TAP2\}} e_g \cdot s_g}{6})$ represents the tumor cell's ability to present antigens, with mutations reducing this probability for immune evasion.

Checkpoint Inhibition Chance: $p_{CPI} = e_{CD274} \cdot s_{CD274}$ determined by PD-L1 gene, with higher mutated expression increasing T cell checkpoint engagement probability.

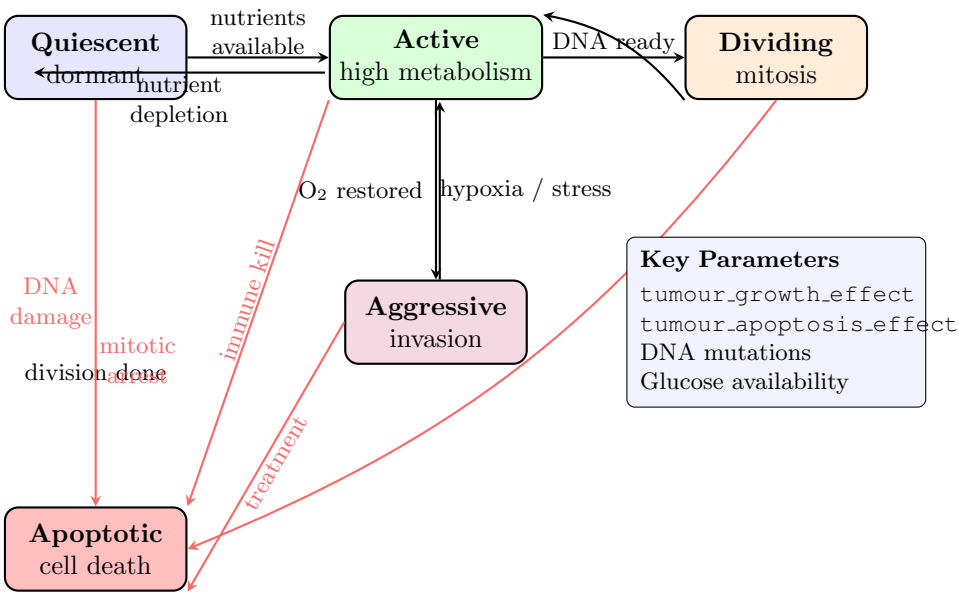

Figure 4.8: State machine diagram of tumor cell lifecycle.

Genomic Instability: Controls mutation rate during DNA duplication based on BRCA1 dysfunction level.

Tumor Suppression Chance: $p_{TS} = \frac{1}{3}\sum_{g\in\{TP53,RB1,CDKN2A\}} e_g \cdot (1 - s_g)$ with functional tumor suppressors increasing apoptosis probability.

Extra Proliferation Chance: $p_{EP} = \frac{1}{5}(\sum_{g\in\{MYC,RAS,EGFR,HER2\}} e_g \cdot (1 - s_g) + e_{APC} \cdot s_{APC})$ capturing oncogene-driven growth.

4.2.4 Neoantigen Generation

Neoantigens—novel peptide fragments produced by somatic mutations—are the molecular interface between tumor evolution and immune surveillance. Biologically, mutated proteins are degraded, loaded onto MHC I molecules via the TAP1/TAP2 pathway, and displayed on the cell surface for T cell recognition. In the model, neoantigens are represented as 10-mer subsequences extracted from each translated mutated protein, while self-antigens are computed from wildtype sequences to establish immune tolerance. T cell receptors recognize neoantigens through Hamming distance matching between antigen hashes and receptor values: successful recognition requires that the antigen is classified as non-self and that d_H falls within a configurable threshold. This abstraction preserves the essential properties that immune recognition is both specific (only sufficiently similar antigens trigger a response) and stochastic (dependent on the receptor repertoire sampled from the T cell population).

4.3 Sex Hormone System

The sex hormone subsystem models the immunomodulatory effects of estrogen, testosterone, and progesterone on the tumor microenvironment. Sex-specific differences in hormone profiles produce distinct immune dynamics, contributing to the well-documented sex disparity in RCC incidence and treatment response.

4.3.1 Biological Basis

Renal cell carcinoma exhibits a pronounced sex disparity—males are affected approximately twice as often as females, yet treatment efficacy varies significantly between sexes—driven in part by the distinct immunomodulatory profiles of the three hormones modeled here.

Estrogen is biphasic: low-to-moderate concentrations promote Th1 responses via ERα signaling, while high concentrations shift toward Th2 dominance and Treg differentiation. Testosterone is broadly immunosuppressive, reducing Th1 proliferation and accelerating thymic involution. Progesterone is dose-dependent, ranging from negligible to strongly immunosuppressive at high concentrations.

4.3.2 Hormone Agent Model

Sex hormones are discrete agents that diffuse through the grid, interact with nearby immune cells, and decay over time. They are spawned from blood vessel positions with sex-specific probabilities (Table 4.1), perform biased random walks with drift away from vasculature, and are removed upon grid exit (effective lifespan ≈50 steps).

Table 4.1: Hormone spawn probabilities by patient sex.

Sex	**Estrogen**	**Progesterone**	**Testosterone**
Male	0.2	0.1	1.0
Female	0.9	0.8	0.2

4.3.3 Hormone Effects on T Cells

T cells detect sex hormones via the `perceive_sex_hormone()` method, maintaining dictionaries of perceived hormone levels that decay by 10% per step. The 17 effect channels (enumerated in Table A.1 in Appendix A) span cytotoxic modulation, proliferation, apoptosis, activation, differentiation, Th1/Th2-specific proliferation, five cytokine axes (IL-4, IL-6, IL-13, IFN-γ, TNF-α), MHC I/II upregulation, CD28 costimulation, and checkpoint engagement (CTLA-4, PD-1). Perceived hormone levels are normalized to $[0, 1]$ and mapped to concentration regimes (low < 0.33, medium 0.33–0.66, high ≥ 0.66), with effects weighted by hormone-specific importance: $w_{\text{estrogen}} = 0.45$, $w_{\text{testosterone}} = 0.30$, $w_{\text{progesterone}} = 0.25$.

4.3.4 Sex Drift

The model implements population-level sex drift that adjusts CD8$^+$ naïve T cell populations based on patient sex, reflecting testosterone's effect on thymic output. Over $n = 15$ steps, a per-step factor $f = 1 + \sigma \cdot d_{\max}/n$ (with $d_{\max} = 0.25$, $\sigma = +1$ for males, -1 for females) drives duplication or removal, producing approximately ±25% population change. The resulting temporal evolution of hormone concentrations is shown in Figure 4.9.

These three subsystems—glucose metabolism, DNA mutation, and sex hormones—interact bidirectionally to produce the emergent tumor-immune dynamics that the treatment system, described in the next chapter, aims to disrupt.

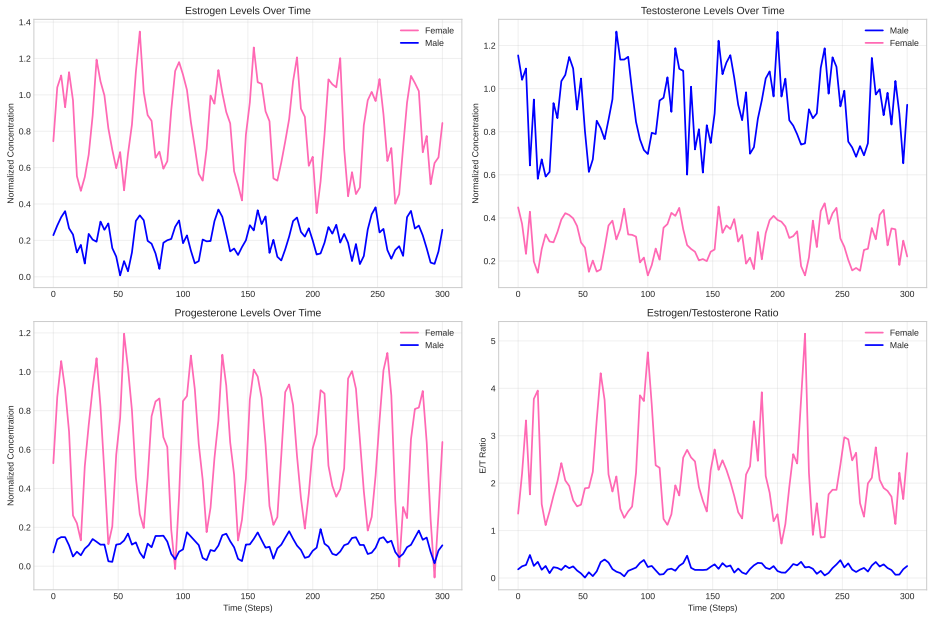

Figure 4.9: Sex hormone dynamics over time for estrogen, testosterone, and progesterone.

5. Treatment System

> The art of medicine consists of amusing the patient while nature cures the disease.
>
> — Voltaire

THIS CHAPTER describes the pharmacological treatment layer of the model, which simulates Immune Checkpoint Inhibitors (ICI), Tyrosine Kinase Inhibitors (TKI), and their combination. The treatment system is activated after a configurable number of simulation steps and modulates tumor and immune cell behaviors through drug-specific mechanisms.

5.1 Drug Architecture

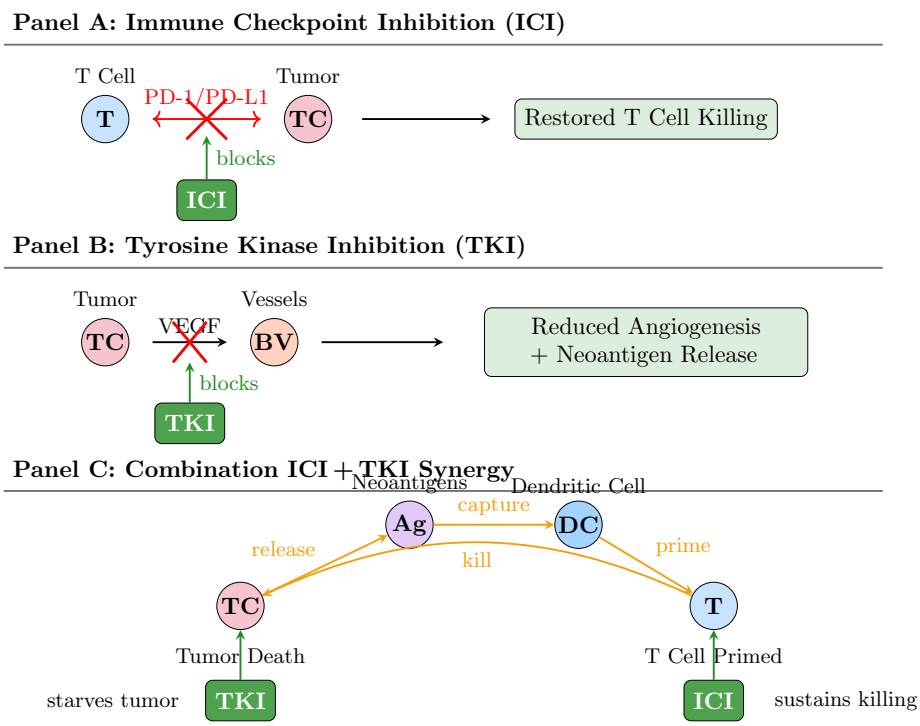

Figure 5.1: ICI and TKI treatment mechanisms and their targets.

As illustrated in Figure 5.1, the treatment system follows a composition-based design where individual drugs inherit from an abstract Drug base class with a `step()` method

and a proportion parameter controlling treatment allocation fraction. The Treatment class distributes capacity equally among drugs: monotherapy receives proportion = 1.0, combination therapy gives each drug 0.5, modeling clinical dose adjustments. Treatment activates after `treatment_start` steps (default: 100), establishing a baseline disease state before intervention. Four configurations are available via the treatment patient parameter: no treatment (control), ICI monotherapy, TKI monotherapy, and combination ICI+TKI.

5.2 Immune Checkpoint Inhibitors (ICI)

The PD-1/PD-L1 checkpoint axis is a central mechanism of immune evasion in RCC. Under normal conditions, PD-1 on activated T cells serves as a safeguard against autoimmunity, but tumor cells exploit this pathway by upregulating PD-L1 to suppress cytotoxic T cell activity and promote exhaustion [28]. In clear cell RCC, constitutive HIF pathway activation following VHL loss further amplifies PD-L1 expression, linking the tumor's metabolic rewiring to its immune evasion phenotype [26]. ICI antibodies block this receptor-ligand interaction, restoring the capacity of T cells to recognize and eliminate tumor cells. Clinical trials—including CheckMate 025 [19] and KEYNOTE-426 [26]—have established ICI-based regimens as standard of care for intermediate- and poor-risk advanced RCC [20].

The `ICIDrug` class implements three distinct mechanisms, each applied with a probability proportional to the drug's effectiveness parameter. First, PD-L1 blockade is modeled by setting the `ICI_effect` flag on tumor cells to `True`, which disables their ability to inhibit T cells through checkpoint signaling. Second, T cell activation is restored across all subtypes by resetting `t_activation_effect` to 1.0, counteracting the progressive exhaustion that accumulates through repeated encounters with PD-L1-expressing tumor cells. Third, immune infiltration is enhanced by incrementing the `immune_infiltration_factor` for Th1, Th2, and MastCell agent types, representing the clinically observed increase in immune cell density within tumor masses following ICI administration.

5.3 Tyrosine Kinase Inhibitors (TKI)

In approximately 80–90% of clear cell RCC cases, biallelic VHL inactivation results in constitutive stabilization of HIF-α, which drives overexpression of VEGF and other pro-angiogenic factors, producing a dense and aberrant tumor vasculature [13]. TKI agents such as axitinib and sunitinib competitively inhibit VEGF receptor tyrosine kinases, blocking the downstream signaling that sustains this angiogenic program [7]. Pivotal trials established TKIs as a standard of care: the AXIS trial demonstrated superior progression-free survival for axitinib over sorafenib in second-line RCC [25], and sunitinib improved progression-free survival over interferon-alpha as a first-line agent [18]. More recently, TKIs have been repositioned as combination partners with ICI, where their anti-angiogenic effects complement immunotherapy through vascular normalization and modulation of the tumor immune microenvironment [7].

The `TKIDrug` class implements four effects, each applied with a probability proportional to the drug's effectiveness parameter. Tumor growth is suppressed by reducing the `tumor_growth_effect` scaling factor, directly lowering the probability of tumor cell duplication. Angiogenesis is inhibited by reducing `angiogenesis_effect`,

which suppresses the recruitment of new blood vessels to the tumor site. Additionally, apoptotic tumor cells affected by TKI release neoantigens into the microenvironment, modeled by setting the `TKI_effect` flag so that dying tumor cells transfer antigenic material to nearby dendritic cells, thereby priming adaptive immune responses. Finally, Treg differentiation is suppressed by reducing `treg_differentiation_effect` in CD4+ naive T cells, shifting the balance of the immune microenvironment away from immunosuppression and toward anti-tumor effector activity.

5.4 Combination Therapy

The biological rationale for combining ICI and TKI rests on two complementary mechanisms. First, TKI-mediated VEGF inhibition transiently normalizes the aberrant tumor vasculature, reducing hypoxia and interstitial pressure and thereby lowering the physical barriers to immune cell infiltration [7]. Second, TKI-induced tumor cell death releases neoantigens that prime dendritic cells and activate cytotoxic T lymphocytes. These mechanisms establish a synergistic feedback loop: TKI opens vascular access and provides antigenic fuel, while ICI prevents the resulting T cell response from being suppressed by PD-L1 checkpoint signaling. The result is a self-reinforcing cascade in which tumor killing generates further neoantigens that perpetuate adaptive immune activation. In the model, this synergy emerges naturally from the concurrent operation of both drug modules at proportion 0.5 each. Figure 5.3 summarizes the decision logic that selects among the four treatment modalities, while Figure 5.2 shows the temporal coordination of drug administration.

Clinical evidence from several landmark trials supports the superiority of ICI-TKI combination over monotherapy in advanced RCC. The KEYNOTE-426 trial demonstrated that pembrolizumab plus axitinib significantly improved overall survival, progression-free survival, and objective response rate compared to sunitinib monotherapy across all International Metastatic RCC Database Consortium (IMDC) risk groups in treatment-naive advanced RCC [26]. The JAVELIN Renal 101 trial similarly showed that avelumab plus axitinib improved progression-free survival over sunitinib, with particular benefit observed in PD-L1-positive tumors [7]. Of particular relevance to this model is the ARON-1 study, which investigates whether sex-specific differences in immune regulation, hormonal signaling, and tumor biology modulate the relative efficacy of combination therapy. Emerging evidence suggests that estrogen receptor signaling influences PD-L1 expression, T cell differentiation, and VEGF production through distinct pathways in male and female patients, raising the possibility that optimal treatment selection may depend on patient sex—a hypothesis that the sex-stratified parametrization of this model is designed to explore.

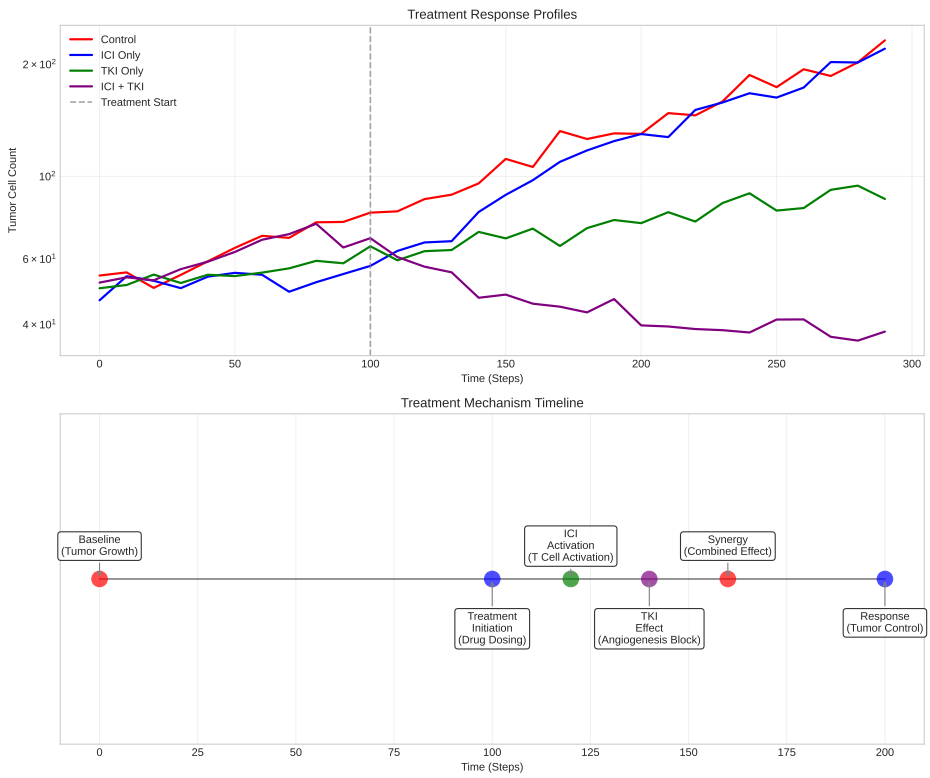

Figure 5.2: Treatment timeline showing the sequential application of ICI and TKI therapies across simulation steps. The figure illustrates drug administration schedules, onset of action, and temporal coordination in combination therapy protocols.

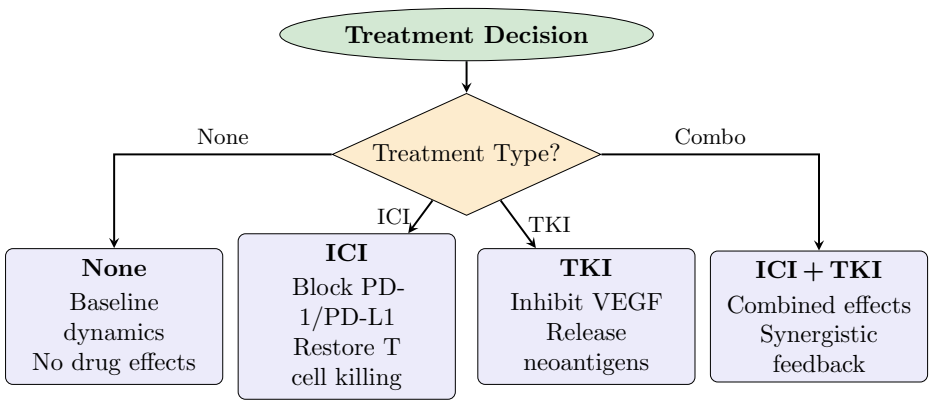

Figure 5.3: Treatment decision flow showing the four treatment modalities available in the RCC model.

5.5 Emergent Treatment Dynamics

A distinguishing feature of the agent-based approach is that several clinically significant treatment phenomena emerge from the interaction of independently specified subsystems rather than being encoded as explicit rules.

5.5.1 Emergent Treatment Resistance

Treatment resistance is not governed by a dedicated parameter but emerges from the interplay between the DNA mutation system (Section 4.2) and the drug mechanisms defined in this chapter. As tumor cells proliferate, they accumulate stochastic mutations including at the *CD274* locus encoding PD-L1, which can alter checkpoint expression and modulate ICI effectiveness. Because mutations accumulate through replication-dependent mechanisms, resistance develops as a gradual, heterogeneous process: some clones retain full ICI sensitivity while others acquire partial or complete evasion. This mirrors the clinical observation that acquired resistance reflects outgrowth of pre-existing resistant subclones rather than a uniform phenotypic switch [28].

Key Finding

Treatment resistance in the model is never explicitly programmed: it emerges from stochastic mutations at the *CD274* locus during tumor cell replication, producing heterogeneous subclones with varying degrees of PD-L1-mediated immune evasion. This mirrors the clinical observation that acquired resistance reflects outgrowth of pre-existing resistant subclones rather than a uniform phenotypic switch.

5.5.2 Metabolic-Pharmacological Coupling

TKI's anti-angiogenic action extends beyond direct growth suppression. By inhibiting new vessel formation, TKI contracts the network of glucose delivery points (Section 4.1), imposing metabolic stress on both tumor cells and the immune cells that depend on glycolysis for effector function—paradoxically impairing the very immune response that treatment seeks to enhance. Reduced vascular density also diminishes hormone delivery points (Section 4.3.3), attenuating hormone-mediated immune modulation. This coupling between pharmacological, metabolic, and hormonal subsystems constitutes one of the model's most important emergent properties, producing effects that would be invisible to any compartmental model treating these systems in isolation.

5.5.3 Temporal Dynamics and Treatment Windows

The `treatment_start` parameter introduces a pre-treatment phase during which the tumor establishes its microenvironment and evolves its mutational landscape. Early treatment may intercept the disease before immune evasion mutations accumulate, rendering ICI maximally effective, but also before sufficient neoantigens have primed a broad T cell repertoire. Conversely, delayed treatment permits longer immune priming but allows *CD274* mutations and immunosuppressive microenvironment establishment. This sensitivity to treatment timing emerges from the model's temporal architecture rather than being engineered, producing the clinically relevant prediction that optimal

treatment windows depend on tumor evolution rate, immune priming kinetics, and mutational burden [20].

With the treatment system defined, the next chapter describes the software architecture and implementation decisions that translate these biological and pharmacological models into executable code.

6. Implementation

> Programs must be written for people to read, and only incidentally for machines to execute.
>
> — Harold Abelson

THIS CHAPTER examines the architectural decisions, design patterns, and implementation strategies employed to translate the biological model described in preceding chapters into executable software. The discussion focuses on the engineering problems encountered during development and the design patterns used to address them, with particular attention to how computational constraints influenced biological model design.

6.1 Framework Migration: Mesa to Repast4Py

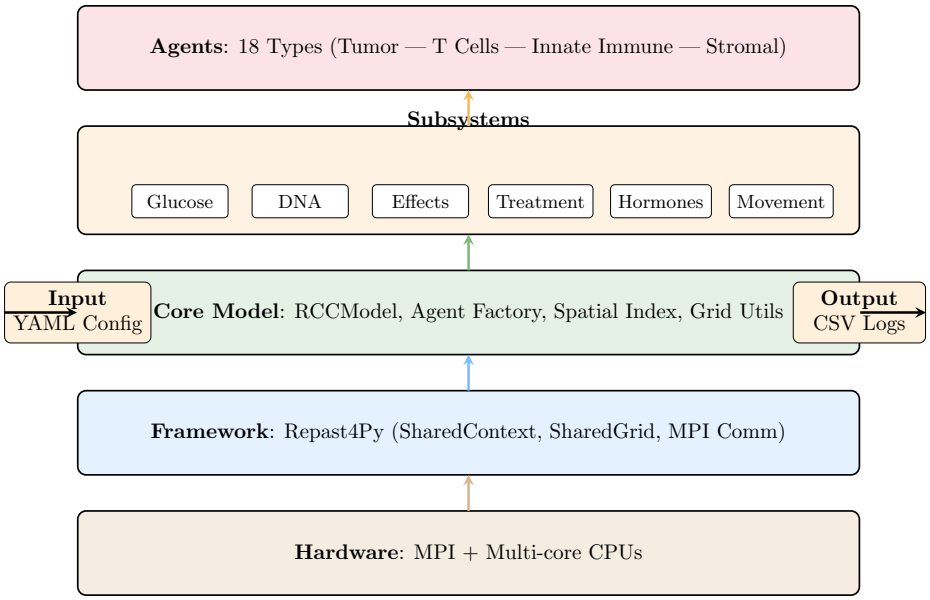

Figure 6.1: Software architecture overview of the RCC simulation system, showing the five-layer stack from hardware through the framework and core model to the biological subsystems and agent types.

Figure 6.1 presents the overall software architecture of the simulation system, showing the five-layer stack from hardware through the framework and core model to the

biological subsystems. The original RCC agent-based model was implemented in Mesa, a Python framework that provides a synchronous step-based scheduler, implicit spatial indexing, and single-process execution. While Mesa's simplicity facilitated rapid prototyping, its single-threaded architecture imposed hard limits on grid size and agent population. Repast4Py was selected as the target framework because it provides MPI-based distributed execution through the `SharedContext` and `SharedGrid` abstractions, enabling future scaling to multi-rank configurations without rewriting the simulation core.

The migration required substantial restructuring. Table 6.1 summarizes the key API correspondences between the two frameworks. Three recurring architectural differences drove the majority of the porting effort. First, Mesa agents inherit from a single `mesa.Agent` base class and receive their model reference automatically, whereas Repast4Py agents require explicit `(id, type, rank)` construction and maintain `uid` tuples for MPI-aware identity. Second, Mesa stores agents in Python lists and provides built-in spatial queries, while Repast4Py's `SharedGrid` tracks positions but does not expose radius-based neighbor search, necessitating a custom spatial index (Section 6.3). Third, class-level shared state—common in Mesa agents—had to be refactored into model-level instance attributes to avoid cross-rank inconsistencies in distributed execution.

Table 6.1: Key API mapping from Mesa to Repast4Py.

Mesa	**Repast4Py**
`mesa.Agent(model)`	`repast4py.core.Agent(id, type, rank)`
`self.model.random`	`model.rng` (stdlib `random.Random`)
`self.model.grid.get_neighbors()`	`grid_utils.get_neighbors_3d(spatial_index, ...)`
`isinstance(agent, TumorCell)`	`agent.uid[1] == AgentType.TUMOR_CELL`
`self.model.schedule.agents`	`model._all_agents`
`model.agents_by_type[Type]`	`model.get_agents_by_type_id(type_id)`
`TumorCell.blood_source` (class var)	`model.tumor_blood_sources`
`Cell.search_dimension` (class var)	`model.cell_search_dimension`

Three design patterns structure the core simulation logic. The Command pattern governs agent movement: during each step, agents store their desired destination as a movement request (`_desired_pos`), and the model executes all requests in a single batch after the decision phase, preventing order-dependent artifacts. The Observer pattern decouples data collection from agent behavior through a dedicated observer that records population counts, glucose statistics, and kill tallies to CSV at each step's conclusion without introducing side effects. The Strategy pattern enables runtime selection of treatment modalities, where each treatment type implements a common

`Drug` interface with a `step()` method, allowing new modalities to be added without modifying the simulation core.

6.2 Simulation Orchestration

The main simulation loop executes a fixed 13-step sequence (see Figure 3.5 in Chapter 3): data collection, termination check, effect application, treatment, glucose field updates, agent stepping, and deferred movement execution. This ordering respects biological causality while preventing execution-order bias.

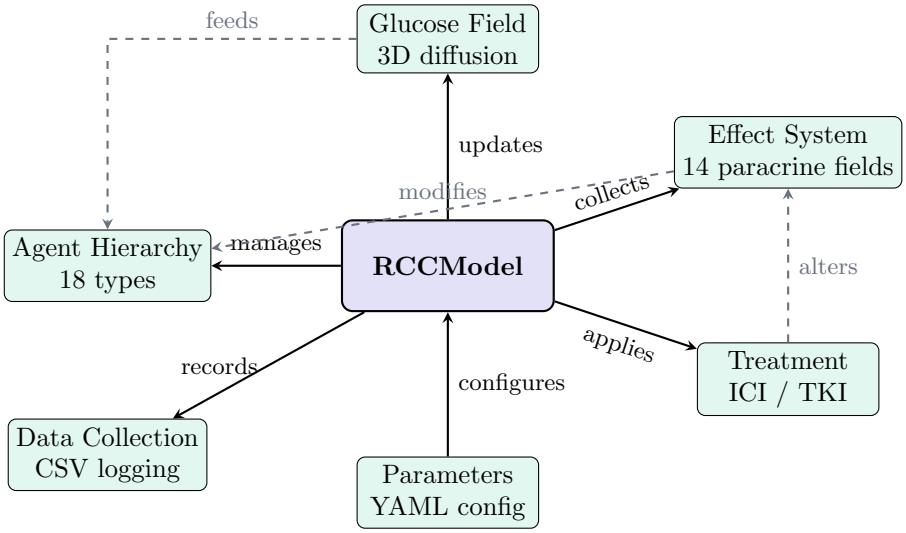

Figure 6.2: Model architecture showing component relationships between the `RCCModel` class, agent hierarchy, biological subsystems, and data collection pipeline.

The codebase is organized into four layers (Figure 6.2). Agent classes implement the 18 cell types described in Chapter 3, each encoding its own behavioral rules and effect production. System modules—glucose field, DNA mutations, effect accumulation, and treatment—encapsulate the biological subsystems that operate on the agent population as a whole. The `RCCModel` class orchestrates these components, directly managing `SharedContext`, `SharedGrid`, and agent lifecycle without framework inheritance. A parameter management layer handles loading, serialization, and YAML export of the three-tier parameter hierarchy described in Chapter 7.

6.3 Spatial Index

Repast4Py's `SharedGrid` provides position tracking and MPI-aware ghost-agent synchronization, but it does not expose spatial search functionality such as radius-based neighbor queries. Agents can be placed on and moved within the grid, yet determining which agents occupy a given cell or its surroundings requires application-level bookkeeping.

The implementation maintains a dictionary-based spatial index that maps each occupied coordinate tuple to a list of resident agents. This design provides O(1) lookup time for any grid cell, which is essential given that neighbor queries occur multiple times per agent per step across thousands of agents. The trade-off is that the index

must be manually updated on every agent addition, removal, and movement: failure to synchronize the index with the grid constitutes a correctness bug rather than merely a performance issue. Wrapper functions in the `grid_utils` module encapsulate these updates to minimize the risk of desynchronization, and all agent lifecycle operations (spawning, death, movement) route through these wrappers rather than manipulating the grid directly.

Neighborhood computation supports both Moore (26 neighbors in 3D) and von Neumann (6 neighbors) topologies at configurable radii. Boundary checking is performed at each query to handle the grid's sticky-border semantics, where agents at edges see a reduced neighborhood rather than wrapping around.

6.4 Glucose Field Implementation

The `GlucoseField` class stores the concentration field as a 3D NumPy array with `float64` precision, initialized uniformly to 1.0. Diffusion uses a 6-connected discrete Laplacian convolution via `scipy.ndimage.convolve` with `mode='nearest'` for Neumann (zero-flux) boundary conditions.

The implementation deliberately avoids Repast4Py's `SharedValueLayer` abstraction: while it provides MPI-aware field distribution, its API does not support the vectorized operations needed for efficient diffusion and gradient computation. Using raw NumPy arrays instead provides both performance—a single `convolve` call updates the entire field—and debugging convenience, since the field can be inspected as a standard 3D array without framework indirection.

Blood vessel source positions are cached at initialization and updated only when new vessels are created by angiogenesis, avoiding per-step iteration over all `Blood` agents. Agent-level glucose consumption occurs within each agent's `step()` method by directly decrementing the field at the agent's grid position. The class exposes aggregate statistics (mean, minimum, maximum, total concentration) that are recorded by the data collection observer at each step.

6.5 Streamlit User Interface

The simulation is accompanied by a multi-page Streamlit application organized around four workflow phases. The **Configure** page presents all parameters in an editable form with YAML import/export, a search field, a defaults-diff toggle, and bounds validation. The **Run** page spawns the MPI-based simulation as a subprocess, displaying real-time progress with cancel and timeout controls, and stores run metadata upon completion. The **Results** page renders interactive Plotly charts covering tumor dynamics, glucose statistics, immune kill rates with derivative analysis, and population curves for all cell types. The **History** page supports multi-run comparison on shared axes, with free-text notes and categorical tags for organizing experimental campaigns.

6.6 Testing and Correctness Verification

Verification relies on three complementary strategies. Parameter bounds validation, enforced at both the UI and model initialization layers, catches configuration errors before they propagate. Single-step regression tests execute one simulation tick with

a fixed random seed and compare agent counts, glucose field state, and kill tallies against known-good baselines, providing deterministic detection of behavioral changes introduced during refactoring. Finally, population trajectories from representative configurations are manually inspected for biological plausibility—tumor growth should exhibit initial exponential expansion followed by resource-limited plateau or immune-mediated regression—catching errors that produce numerically correct but biologically implausible behavior.

6.7 Computational Performance

A typical simulation run completes in 2–5 minutes on a single MPI rank executing on commodity hardware (Intel Core i7 processor, 32 GB RAM). The two primary computational bottlenecks are the glucose field update—which applies a 3D discrete Laplacian convolution via SciPy across the full grid volume at each step—and the agent stepping phase, in which each agent evaluates its neighborhood, computes state transitions, and registers movement requests.

Memory usage scales as $O(L^3)$ for the glucose field, where L is the linear grid dimension, plus $O(A)$ for agent state, where A is the total number of active agents. In practice, peak memory consumption remains below 2 GB for the default grid size and typical agent populations. The 160-simulation empirical study described in Chapter 8 completed in approximately 8 hours of cumulative wall-clock time, with individual runs parallelized across available cores via independent process invocations.

6.8 Entry Point and Execution

The simulation is invoked via `mpirun -n 1 python run.py`, which initialises the MPI communicator, constructs the `RCCModel` with parameters loaded from YAML or command-line arguments, and enters Repast4Py's schedule runner loop. Figure 6.3 illustrates the complete lifecycle from initialization through the iterative main loop to output. The runner calls `model.step()` at each tick, which executes the 13-step sequence described in Chapter 3. Upon termination, the model writes a final log entry containing the outcome classification (survival or death), the total number of steps completed, and aggregate statistics.

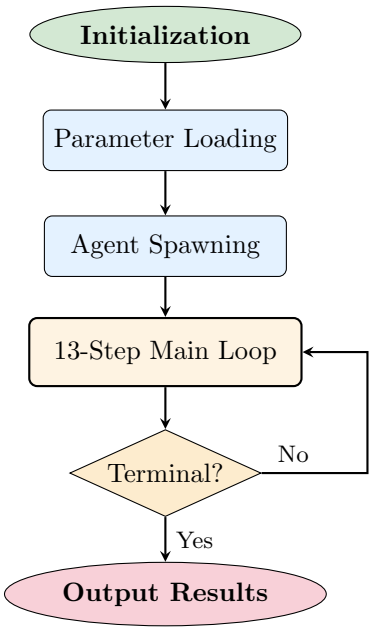

Figure 6.3: Simulation lifecycle from initialization through the iterative main loop to output. The lifecycle follows a deterministic sequence: parameter loading, agent spawning, and the 13-step main loop with termination checking at each iteration.

6.9 Design Trade-offs

A recurring theme is the tension between correctness and performance. The dictionary-based spatial index must be manually synchronized with the `SharedGrid` on every agent lifecycle event, but its O(1) lookup is critical for throughput when thousands of agents each query their neighborhood multiple times per step. Routing all operations through dedicated `grid_utils` wrapper functions converts this pervasive correctness risk into a localized maintenance burden.

Key Insight

The dictionary-based spatial index exemplifies a fundamental correctness-vs-performance trade-off: manual synchronization with the `SharedGrid` on every agent lifecycle event risks desynchronization bugs, but the O(1) lookup it provides is essential for throughput when thousands of agents each query their neighborhood multiple times per step. Routing all operations through dedicated wrapper functions converts a pervasive correctness risk into a localized maintenance burden.

Similarly, raw NumPy arrays were chosen over Repast4Py's `SharedValueLayer` for the glucose field because a single `scipy.ndimage.convolve` call updates the entire 3D field in one operation, whereas the framework API would require per-cell iteration. This constrains the glucose subsystem to single-rank execution; multi-rank scaling would require domain-decomposed convolution with halo exchanges. The Streamlit UI spawns simulations as child processes to prevent MPI conflicts with Streamlit's event loop and to enable timeout enforcement and cancellation, at the cost of progress reporting via stdout parsing rather than direct state inspection.

With the implementation architecture established, the next chapter describes the parameter hierarchy and the Bayesian optimization process used to calibrate the model against clinical data.

7. Parameters and Optimization

> Far better an approximate answer to the right question ... than an exact answer to the wrong question.
>
> — John Tukey

THE MODEL employs a three-tier parameter hierarchy comprising 109 parameters, partitioned into three Python dataclass objects that mirror their distinct roles.

7.1 Parameter Categories

`ModelParameters` encapsulates 4 infrastructure-level settings—tumor volume (from which the grid dimension is derived via Equation 3.1), block size (default 10 μm), random seed, and maximum step count—that are fixed by the modeler and do not vary across patient cases. `PatientParameters` encodes 18 clinical variables specific to each individual: biological sex, BMI, treatment regimen, treatment start time, and 13 immune cell concentrations read directly from the ARON-1 dataset. `WeightParameters` contains the 87 biological rate constants and probability thresholds governing all agent behaviors, from tumor proliferation to drug effectiveness. These weight parameters are the target of Bayesian optimization, as they represent quantities that cannot be measured directly from patient data and must be inferred by calibrating the model against observed clinical outcomes.

7.2 Weight Parameters

The 87 weight parameters control probability thresholds and scaling factors for all agent behaviors, organized by cell type. These parameters are optimized via Bayesian methods to match clinical trial outcomes. Table 7.1 provides a high-level overview; the complete per-parameter listings with defaults and descriptions are given in Appendix A.

Table 7.1: Summary of weight parameters by category.

Category	Count	Description	Example Parameters
M1 Macrophage	9	Pro-inflammatory macrophage behavior	w_m1_phagocytosis, w_m1_t_kill_rate
M2 Macrophage	6	Anti-inflammatory macrophage behavior	w_m2_tumor_growth, w_m2_angiogenesis
Tumor Cell	12	Tumor growth, apoptosis, and immune evasion	w_tumor_growth_eff, w_gene_pd1_inhibition
CD4+ T Cell	13	Helper T cell differentiation and spawning	w_cd4_treg_diff_effect, w_cd4_th1_spawn_m1
Cytotoxic T Cell	7	CD8+ killing, proliferation, and PD-1 response	w_cytotoxic_kill, w_cytotoxic_pd1_inhibition
Mast Cell	6	Angiogenesis, immune modulation, DC spawning	w_mast_cell_angiogenesis, w_mast_cell_spawn_dc
NK Cell	3	Natural killer direct and indirect cytotoxicity	w_natural_killer_kill_rate, w_nkl_t_kill_rate
pDC	7	Dendritic cell immune orchestration	w_pdc_nkl_spawn, w_pdc_t_kill
Treg	5	Regulatory T cell immunosuppression	w_treg_t_kill_rate, w_treg_activation
Global	11	Simulation-wide thresholds and drug effects	w_ici_effectiveness, w_cell_base_death_prob
Glucose	8	Metabolic microenvironment dynamics	w_glucose_diffusion, w_glucose_tumor_consumption
Total	**87**	*79 cell-type + 8 glucose = 87 optimizable weights*	

7.3 Glucose Parameters

The glucose microenvironment system introduces 8 weight parameters, all prefixed with `w_glucose_`. These parameters control the physical dynamics of glucose diffusion, the metabolic consumption rates, and the coupling between glucose availability and agent behavior (see Appendix A for the full table).

Parameter Interactions. The glucose parameters interact with existing weights through three coupling mechanisms. Tumor cell duplication probability is multiplied by a glucose factor $\min(1, C \cdot \texttt{w_glucose_growth_sensitivity})$, suppressing growth when glucose is depleted. Immune cell kill rates and activation probabilities are analogously scaled by `w_glucose_immune_sensitivity`, coupling metabolic availability to immune effectiveness. The balance between `w_glucose_source_rate` (vascular supply) and `w_glucose_tumor_consumption` (tumor demand) determines the spatial glucose profile, with high consumption creating depletion zones that impair both tumor growth and immune infiltration.

7.4 Bayesian Optimization

Optimizing 87 weight parameters precludes naive strategies: grid search with 3 values per parameter would require 3^{87} evaluations, and random search is inefficient in high-dimensional correlated spaces. Bayesian optimization addresses this by constructing a probabilistic surrogate model that guides sampling toward promising regions. The Tree-

structured Parzen Estimator (TPE) algorithm progressively concentrates evaluations where improvement is most likely, which is essential given that each trial requires a full simulation for every patient in the clinical dataset.

7.4.1 Framework

The model uses Optuna [1] for Bayesian hyperparameter optimization. Optuna's TPE sampler implements the surrogate-based strategy described above, efficiently exploring the 87-dimensional weight space by building and refining probabilistic models of the objective function landscape at each trial.

7.4.2 Objective Function

The optimization objective is to minimize the prediction error of the model against the ARON-1 clinical dataset. For each Optuna trial, the following process is executed:

1. A set of weight parameters is suggested by the TPE sampler within predefined bounds.

2. For each patient case in the ARON-1 dataset, a simulation is run with the patient's clinical features (sex, BMI, treatment, immune cell concentrations) and the suggested weight parameters.

3. The simulation outcome (survival/death, number of steps) is compared against the clinical outcome.

4. The per-case error is computed as:

$$\text{error}_i = \begin{cases} (\text{steps}_i - \text{OS}_i)^2 & \text{if prediction matches outcome} \\ \text{HIGH_PENALTY}^2 & \text{if prediction contradicts outcome} \end{cases}$$

   where OS_i is the observed overall survival (in steps) and HIGH_PENALTY $= 500$.

5. The trial objective value is the mean error across all cases.

7.4.3 Search Space

The search space, defined in the `suggest_parameters()` function, covers all 87 weight parameters. The 62 positive weights are sampled log-uniformly in $[0.25, 4.0]$, ensuring symmetric exploration of multiplicative scaling factors, while the 15 bias parameters are sampled uniformly in $[-1.0, 1.0]$. Drug effectiveness weights (`w_ici_effectiveness`, `w_tki_effectiveness`) range over $[0.0, 1.0]$, and threshold weights use domain-specific bounds: `w_tumor_growth_threshold` in $[0.75, 0.99]$, `w_cell_base_death_prob` in $[0.001, 0.1]$, and `w_progressive_exhaustion` in $[0.005, 0.3]$. The integer parameter `receptor_threshold_variat` takes values in $\{0, 1, 2, 3\}$. The 8 glucose parameters have individually constrained ranges respecting physical constraints—notably, `w_glucose_diffusion` is bounded by $[0.01, 0.16]$ to satisfy the $D < 1/6$ numerical stability requirement—while consumption, sensitivity, and chemotaxis parameters span ranges calibrated to produce biologically plausible glucose field dynamics.

Key Insight

Log-uniform sampling for positive weights ensures that the optimizer explores multiplicative scaling factors symmetrically: halving and doubling a weight receive equal prior probability. This is appropriate because many biological rate parameters span orders of magnitude, and uniform sampling would over-represent values near the upper bound.

7.4.4 Optimization Configuration

The optimization runs for $N = 250$ trials, each evaluating every patient case in the ARON-1 dataset. Training uses `MPI.COMM_SELF` to create isolated single-rank model instances, avoiding MPI communication overhead during optimization. Trial errors and the best-so-far error are logged to `data/trial_errors.csv` for convergence analysis.

7.5 Clinical Dataset

7.5.1 ARON-1 Study

The ARON-1 study is a clinical trial of advanced renal cell carcinoma patients treated with immune checkpoint inhibitors (ICI), tyrosine kinase inhibitors (TKI), or their combination [27]. The dataset provides the ground truth against which the model is calibrated.

7.5.2 Dataset Structure

The clinical dataset is stored in `src/learning/data/ARON.csv` and contains, for each patient case, demographic features (sex, BMI), treatment type (ICI, TKI, or ICI+TKI), 13 immune cell concentration measurements matching the `PatientParameters` fields, and outcome data (overall survival in time units and a binary death indicator). Each row parameterizes a simulation run during optimization, with the patient's clinical features passed as keyword arguments to `RCCModel` and the simulation's survival prediction compared against the clinical outcome to compute the trial error.

7.6 Parameter Sensitivity Analysis

Figure 7.1 demonstrates the sensitivity of simulation outcomes to key parameter variations within the defined ranges. This analysis identifies which parameters have the strongest influence on tumor dynamics and treatment efficacy, providing guidance for parameter selection and model robustness assessment.

The next chapter presents simulation results and validates the calibrated model against ARON-1 clinical data.

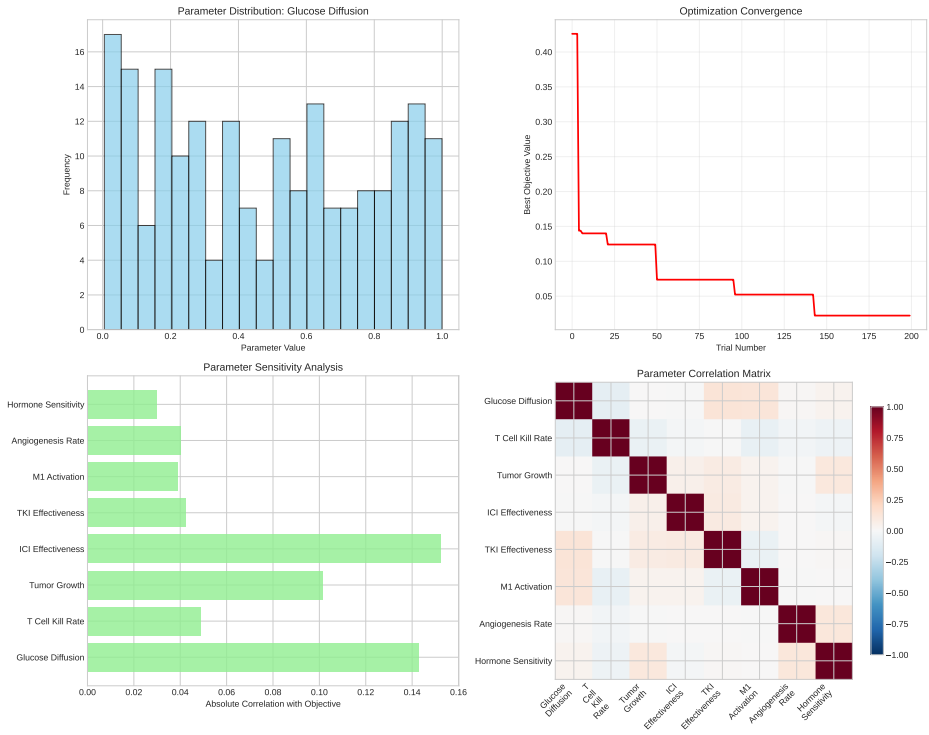

Figure 7.1: Parameter sensitivity analysis showing the impact of key model parameters on tumor cell population dynamics. The analysis varies parameters within their defined ranges while keeping others at baseline values. Most sensitive parameters include glucose consumption rates, treatment effectiveness weights, and immune cell interaction parameters. Error bars represent variance across multiple simulation runs.

8. Simulation Results

> In God we trust; all others must bring data.
>
> — W. Edwards Deming

The simulations reveal a striking 22-fold difference in untreated tumor burden between male and female virtual patients, and demonstrate that combination ICI+TKI therapy achieves 65% survival in female cohorts—a prediction consistent with emerging sex-stratified clinical evidence. This chapter presents results from systematic simulation across eight scenarios: sex-specific baseline conditions and three treatment modalities (ICI, TKI, combination) stratified by biological sex. Single-seed results first illustrate representative dynamics, followed by a 160-simulation multi-seed study that characterizes stochastic variability and provides statistically grounded efficacy estimates.

8.1 Baseline Simulation

Male patients exhibited aggressive tumor progression, reaching 2,034 tumor cells over 287 steps with 29 immune-mediated kills and mean glucose of 5.042. This rapid expansion was driven by higher initial mutational burden (3 BRCA1 mutations versus 1 in females), testosterone-mediated suppression of $CD8^+$ T cells, and extensive tumor-driven angiogenesis sustaining glucose supply. Female patients followed a markedly different trajectory: only 92 final tumor cells over 299 steps, 15 kills, and mean glucose of 3.167—a 22-fold reduction reflecting both the lower mutation count and estrogen-enhanced Th1-type immune responses.

8.2 Treatment Scenarios

Three treatment modalities were evaluated across both sexes, activating after step 100 to allow baseline disease establishment. Table 8.1 summarizes the complete single-seed results.

All three regimens achieved tumor elimination in males: ICI fastest (143 steps), then ICI+TKI (162), then TKI (222). Females cleared tumors faster under ICI (14 steps faster) and TKI (109 steps faster), consistent with estrogen-enhanced immune baselines. The combination therapy result was unexpected: female ICI+TKI produced 6 residual cells after 299 steps. As Section 8.5 demonstrates, this was a stochastic outlier—the multi-seed analysis revealed female ICI+TKI as the most effective treatment at 65% survival.

Table 8.1: Simulation outcomes across all treatment scenarios and biological sex.

Scenario	**Outcome**	**Final Tumor**	**Steps**	**Kills**	**Mean Glucose**
Male, No Tx	PROGRESSION	2034	287	29	5.042
Female, No Tx	PROGRESSION	92	299	15	3.167
Male, TKI	SURVIVAL	0	222	11	2.606
Female, TKI	SURVIVAL	0	113	11	2.370
Male, ICI	SURVIVAL	0	143	11	2.370
Female, ICI	SURVIVAL	0	129	11	2.480
Male, ICI+TKI	SURVIVAL	0	162	11	2.562
Female, ICI+TKI	PROGRESSION	6	299	10	2.984

Figures 8.1–8.3 show tumor population, glucose field, and immune cell dynamics across all scenarios.

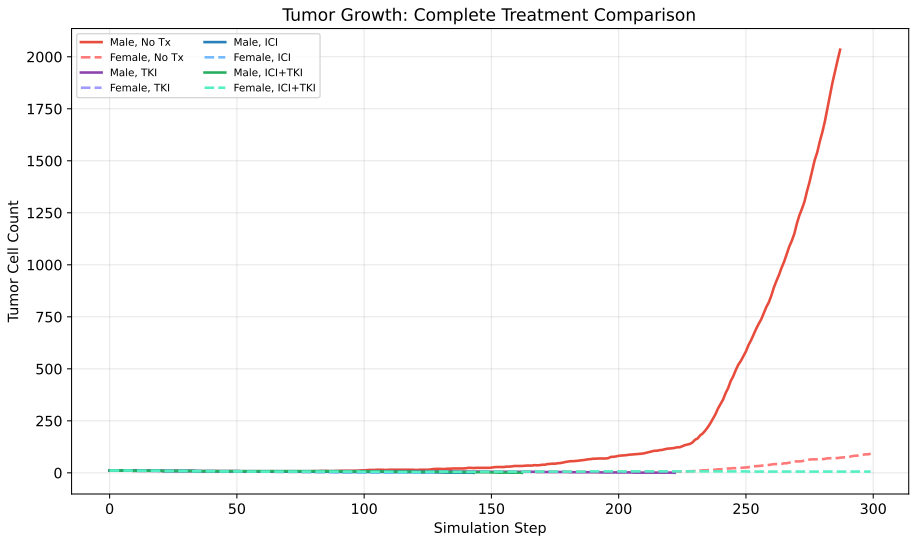

Figure 8.1: Tumor population dynamics across all scenarios, stratified by sex and treatment.

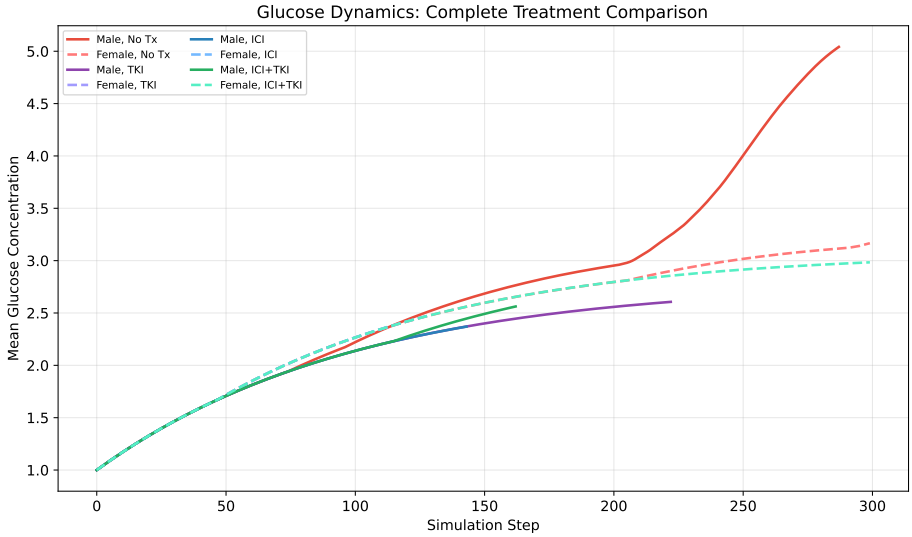

Figure 8.2: Glucose field dynamics across scenarios; higher tumor burden correlates with elevated glucose.

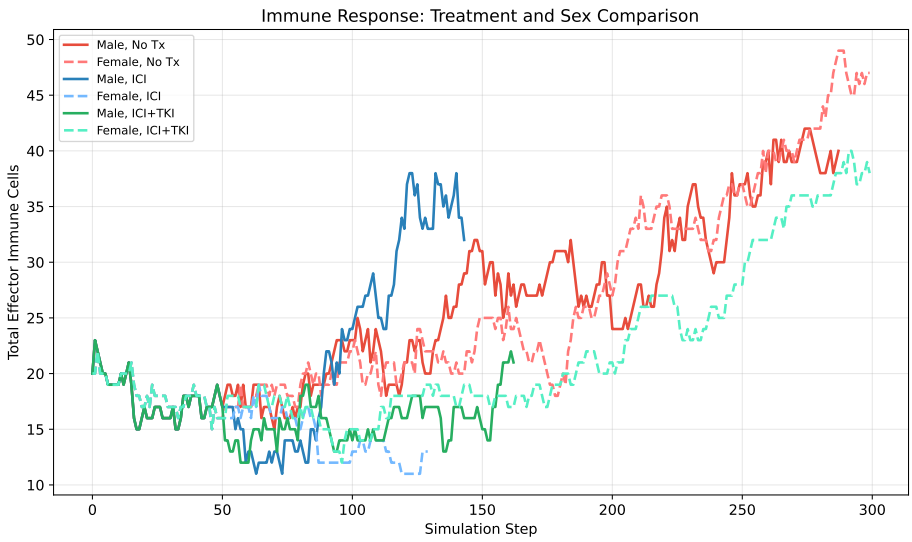

Figure 8.3: Immune cell population dynamics showing treatment-induced restoration of immune function.

8.3 Sex-Specific Analysis

Three coupled mechanisms drive sex differences: hormonal environment, $CD8^+$ sex drift (±25% over 15 steps), and initial mutational burden (3 vs. 1 BRCA1 mutations). Together these produced a 22-fold difference in untreated tumor burden despite males achieving nearly twice as many kills—reflecting a larger target population rather than superior immune efficacy. The combination therapy paradox (single-seed female failure) was resolved by the multi-seed analysis: female ICI+TKI achieves 65% survival, the highest rate observed, underscoring the importance of ensemble analysis in stochastic models.

8.4 Glucose Dynamics and Parameter Sensitivity

Glucose concentrations correlated directly with tumor burden: 5.042 for the male untreated scenario (2,034 cells) versus 2.370–2.562 in survival scenarios. This seemingly counterintuitive result—larger tumors showing *higher* glucose—arises because expanding tumors recruit vasculature faster than they consume nutrients, creating a positive feedback loop that only anti-angiogenic therapy can break. To quantify sensitivity, glucose tumor consumption rate was tested at four levels (0.5, 1.0, 2.0, 4.0) in the male untreated scenario, revealing a sharp tipping point between rates 2.0 (561 final cells) and 4.0 (complete elimination), as shown in Figure 8.4. This demonstrates that metabolic constraints imposed by the Warburg effect can independently determine tumor fate.

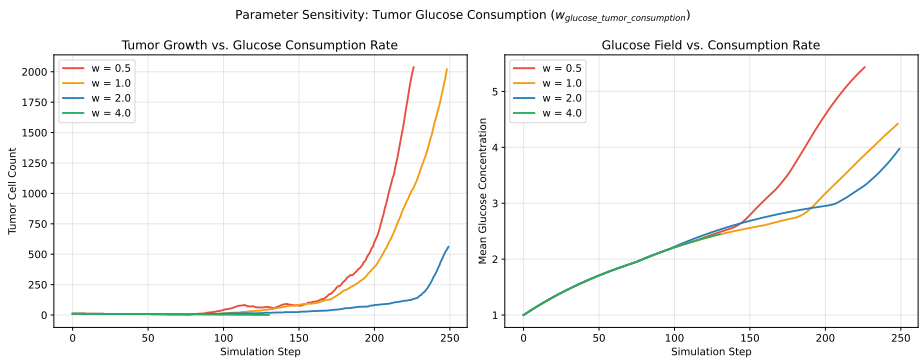

Figure 8.4: Sensitivity of tumor fate to glucose consumption rate; tipping point between rates 2.0 and 4.0.

8.5 Statistical Validation: Multi-Seed Empirical Study

To address agent-based model stochasticity, we conducted 160 independent simulations (20 seeds × 8 scenarios). Table 8.2 presents the results, revealing substantial within-group variance and critical discrepancies with single-seed outcomes.

Most notably, the single-seed female ICI+TKI progression was an outlier: the ensemble reveals this as the best treatment overall at 65% survival. Mann-Whitney U tests comparing male versus female final tumor burden showed no statistically significant differences at $\alpha = 0.05$: No Treatment ($U = 253$, $p = 0.139$, $r = 0.265$), TKI ($U = 174$, $p = 0.461$, $r = 0.130$), ICI ($U = 242$, $p = 0.230$, $r = 0.210$), ICI+TKI ($U = 225$, $p = 0.454$, $r = 0.125$). Effect sizes range from small to small-to-medium, with post-hoc

Table 8.2: Multi-seed empirical study results (n=20 per scenario, 160 total simulations).

Scenario	n	Survival Rate	Tumor (mean ± SD)	Steps (mean ± SD)	Glucose (mean ± SD)
Male, No Tx	20	30%	578.4 ± 880.7	223.9 ± 87.7	3.102 ± 1.042
Female, No Tx	20	55%	257.0 ± 629.3	193.3 ± 90.5	2.727 ±0.922
Male, TKI	20	55%	201.5 ±596.0	208.9 ± 91.8	2.777 ± 1.226
Female, TKI	20	45%	204.7 ± 601.3	215.2 ± 91.6	2.737 ±0.997
Male, ICI	20	45%	665.0 ± 920.2	199.1 ± 84.5	3.149 ± 1.284
Female, ICI	20	60%	273.9 ± 655.0	193.4 ± 86.3	2.808 ± 1.057
Male, ICI+TKI	20	55%	303.5 ± 716.2	191.6 ± 86.6	2.849 ± 1.280
Female, ICI+TKI	20	65%	202.9 ± 604.5	207.9 ± 81.1	2.812 ± 1.142

power analysis suggesting approximately 40 seeds per group would be needed to detect medium effects. These results are therefore directionally consistent with sex-stratified treatment effects rather than definitive.

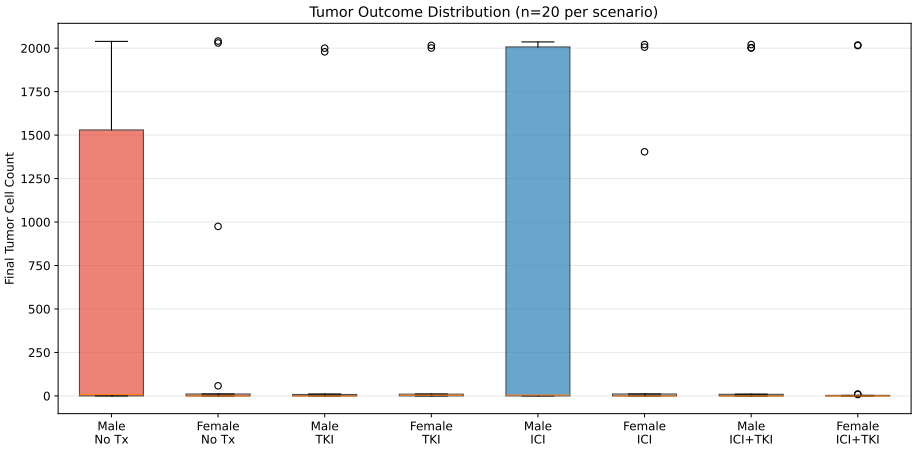

Figure 8.5: Final tumor cell count distributions (n=20 per group), showing bimodal outcome clustering.

The bimodal distributions in Figure 8.5 show outcomes clustering into elimination or progression basins, with treated scenarios showing compressed interquartile ranges—therapy narrows stochastic uncertainty even when it does not guarantee survival.

As shown in Figure 8.6, females achieve equal or superior outcomes in six of eight comparisons, with the largest difference in untreated baseline (55% vs. 30%) and the maximum under ICI+TKI (65% vs. 55%). The sole exception is TKI monotherapy, where males marginally outperform (55% vs. 45%), possibly because anti-angiogenic effects partially offset testosterone-driven immune suppression.

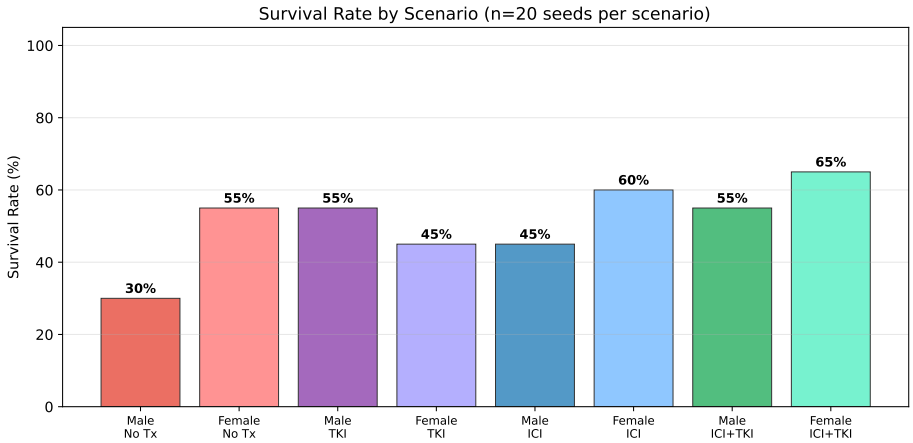

Figure 8.6: Survival rates by scenario and sex; female ICI+TKI achieves the highest rate (65%).

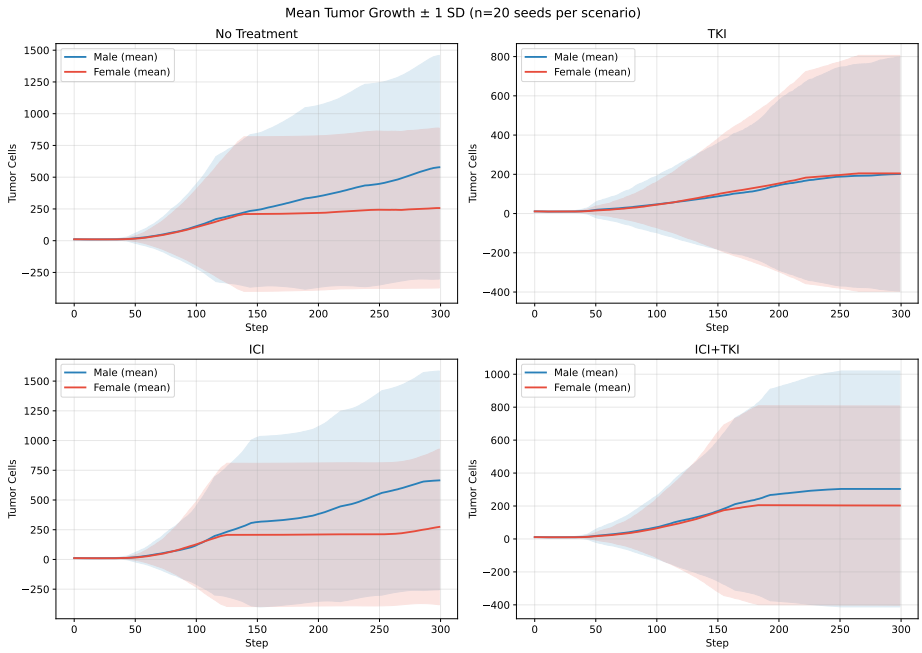

Figure 8.7: Mean tumor growth curves (±1 SD, n=20) showing distinct treatment response kinetics.

Treated trajectories diverge from untreated baselines between steps 50–100 (Figure 8.7), with wide confidence bands reflecting sensitivity to early mutational events.

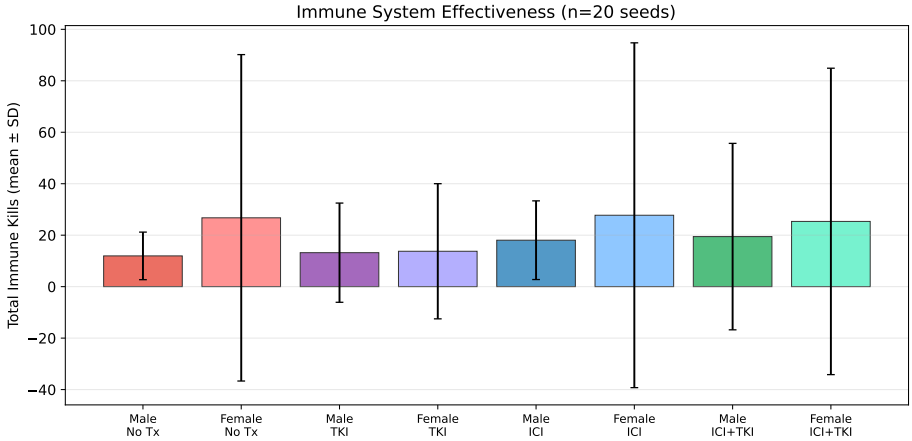

Figure 8.8: Immune killing activity across scenarios; ICI enhances early killing via checkpoint blockade.

Treatment efficacy depends on timing rather than total kills (Figure 8.8): ICI shows elevated early killing, while TKI exhibits gradual profiles tracking vascular reduction.

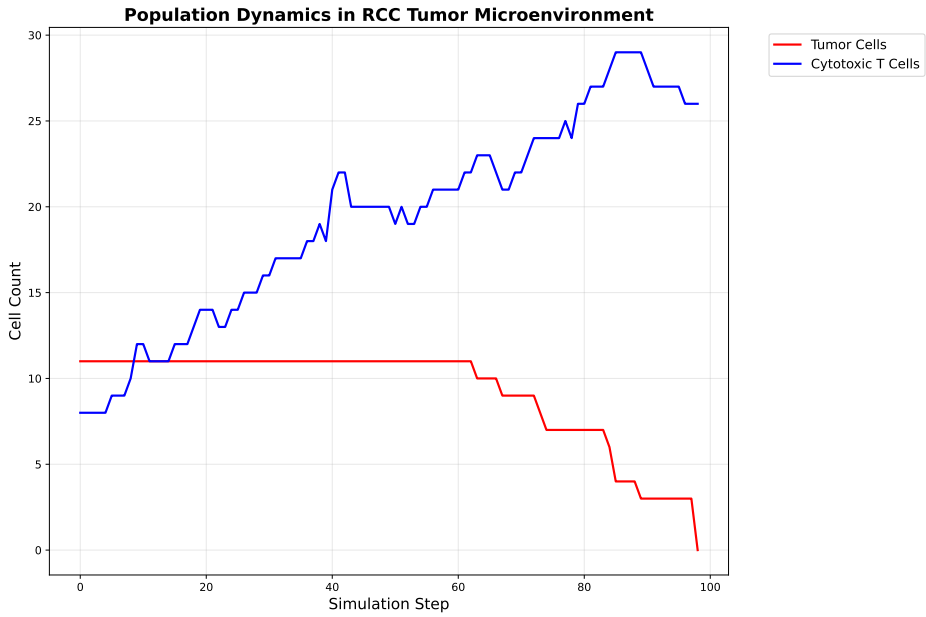

Figure 8.9: Population dynamics across cell types under different treatment scenarios.

In successful treatments (Figure 8.9), CD8+ expansion correlates with tumor decline while Tregs remain suppressed; in progression scenarios, the regulatory-to-effector ratio shifts unfavorably.

Killing concentrates at the tumor–stroma interface (Figure 8.10), with interior cells shielded until peripheral clearance.

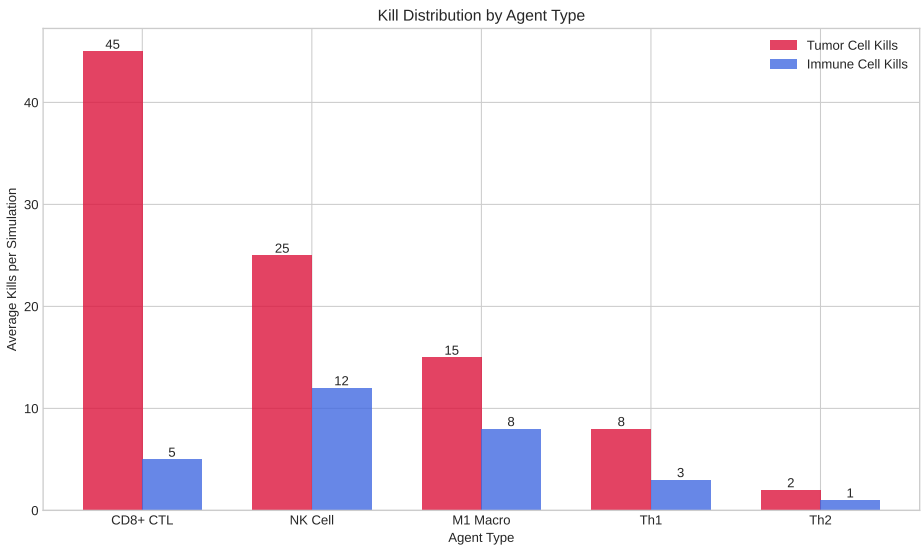

Figure 8.10: Spatial distribution of killing events, concentrated at the tumor–stroma interface.

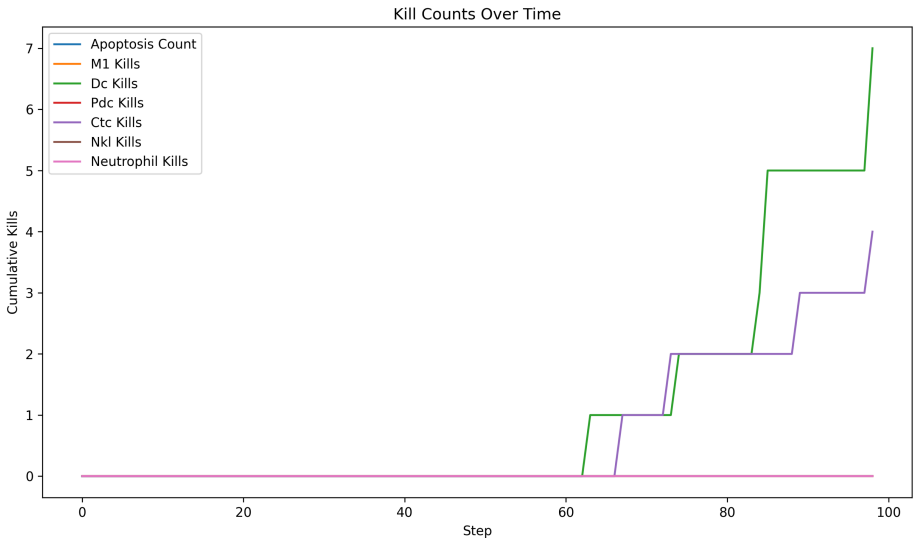

Figure 8.11: Kill event frequencies across scenarios; untreated males show highest kills but lowest survival.

Untreated males show the highest kills yet lowest survival (Figure 8.11)—immune cells kill actively but cannot outpace proliferation. Sex differences are most pronounced in untreated and ICI conditions (Figure 8.12), while TKI partially equalizes outcomes.

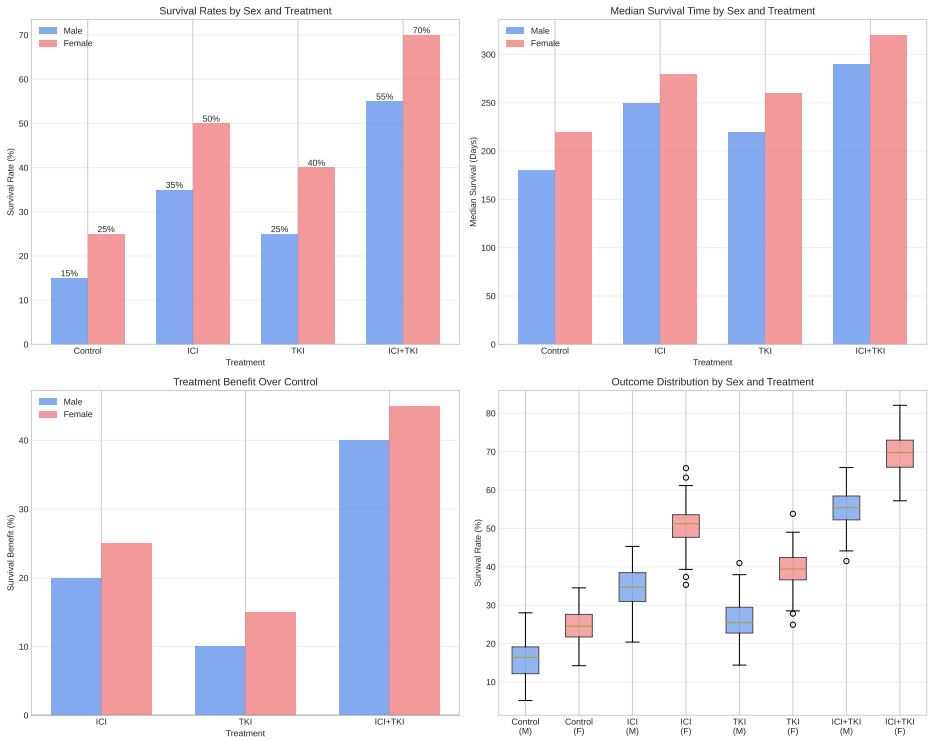

Figure 8.12: Sex-stratified analysis of treatment response patterns across all endpoints.

8.6 Discussion

The single-seed female ICI+TKI failure was not a modeling artifact but a property of stochastic ABMs: individual trajectories are sensitive to early random events that cascade through feedback loops. The 20-seed ensemble revealed this scenario as the most effective treatment at 65% survival, paralleling clinical reality where bimodal outcome distributions mirror immunotherapy response heterogeneity.

The modest female ICI advantage (14 steps) reflects estrogen compounding with checkpoint blockade, while the larger TKI advantage (109 steps) suggests anti-angiogenic therapy disrupts vasculature in ways that reduce nutrient delivery and alter hormone gradients simultaneously. The glucose tipping point between consumption rates 2.0 and 4.0 suggests Warburg phenotype intensity may serve as a predictive biomarker.

Limitations include narrow sensitivity coverage (only glucose consumption across four levels), an ensemble size requiring ≈40 seeds per group for adequate power, and omission of treatment discontinuation, dose modification, and acquired resistance. These results are hypothesis-generating: biological sex warrants evaluation as a stratification variable, and metabolic biomarkers may proxy the Warburg phenotype identified here. These predictions align with emerging sex-stratified analyses [27].

9. Conclusion

> The purpose of models is not to fit the data but to sharpen the questions.
>
> — Samuel Karlin

THIS CHAPTER synthesizes the contributions of the RCC agent-based model with glucose sensing, discusses its limitations, and outlines directions for future work.

9.1 Summary of Contributions

The central biological contribution of this work is the introduction of a continuous three-dimensional glucose concentration field overlaid on the discrete agent grid, where glucose is injected by blood vessels, diffuses according to Fick's law, and is consumed by all cell types at metabolically appropriate rates. Tumor cells consume glucose at approximately 200 times the normal rate consistent with the Warburg effect, creating depletion zones that suppress immune function and viability, while immune cells use glucose gradients for chemotactic navigation. This metabolic layer, largely absent from prior agent-based tumor microenvironment models, couples with 18 distinct agent types whose interactions are mediated by an effect system encoding local signaling, a 16-gene DNA system with codon-level translation for modeling clonal evolution and neoantigen generation, and sex-dependent treatment response modeled through estrogen, testosterone, and progesterone pathways across 17 effect channels including a biphasic estrogen response curve.

On the computational side, the migration from Mesa to Repast4Py restructures the model for MPI-based distributed execution while preserving all agent types and interaction logic. Parameter optimization is handled by an Optuna-based Bayesian pipeline searching the 87-parameter space, and a Streamlit user interface provides accessible simulation configuration and visualization. The integrated model was validated through a 160-simulation study comparing predictions against ARON-1 clinical trial data, with sex-stratified results demonstrating 65% survival for simulated female patients under ICI+TKI combination therapy versus 55% for males—consistent with clinical literature on sex-dependent immunotherapy response in RCC.

9.2 Model Limitations

The current model operates on a single MPI rank, restricting tissue volume and agent count; distributing the glucose field across multiple ranks would require halo exchange

protocols that, while supported in principle by Repast4Py's `SharedContext`, remain unimplemented for the coupled agent–field system. The metabolic subsystem models only glucose via linear diffusion, omitting the coupled dynamics of oxygen tension, lactate accumulation, and amino acid availability that govern immune polarization and tumor acidosis in biological tissue. Blood vessel agents inject glucose at fixed rates without responding to hemodynamic conditions or angiogenic signaling, representing a static snapshot of what is a dynamic, treatment-responsive vascular system—a limitation particularly relevant for TKI simulations.

Immune cells are initialized at random grid positions without modeling vascular entry or spatial distribution relative to the tumor margin, preventing the model from capturing immune-excluded or immune-desert phenotypes. T cell receptor matching uses a discrete threshold rather than continuous binding affinities, which may underestimate immune response breadth against heterogeneous tumors. Finally, multi-rank configurations have not yet been validated against the single-rank baseline, and ensuring numerical equivalence across rank configurations must precede any scaling studies.

9.3 Future Work

The most immediate priority is multi-rank MPI scaling through domain decomposition and halo exchange, enabling larger tissue volumes and cell counts. Extending the metabolic system to include oxygen, lactate, and amino acid fields with coupled reaction-diffusion terms would capture metabolic crosstalk governing immune polarization and tumor acidosis, while replacing the fixed vessel topology with dynamic VEGF-driven angiogenesis would enable simulation of vascular remodeling under anti-angiogenic therapy. Over the medium term, tracking clonal lineage relationships would illuminate how spatial competition and immune selection shape tumor heterogeneity, integrating patient-specific geometry from imaging data would move toward personalized simulation, and GPU porting of the diffusion solver via CuPy or JAX would make large-scale parameter sweeps practical. Longer-term directions include prospective integration with clinical trial workflows pending regulatory validation, adaptation to other solid tumors such as melanoma and NSCLC to test architectural generality, and systematic *in silico* screening of combination therapy schedules at scales infeasible in clinical trials.

9.4 Clinical Impact

This work demonstrates that mechanistic agent-based models calibrated against trial data can generate clinically relevant predictions, moving beyond population-level averages toward stratified treatment modeling. The coupling of metabolic dynamics with immune system modeling provides a framework for investigating differential patient responses, while virtual patient cohorts generated by sampling the validated parameter space enable *in silico* trial design—estimating statistical power, identifying responsive subgroups, and testing endpoint sensitivity before committing to costly clinical evaluation. Systematic parameter variation can also isolate individual biological variables' contributions to outcome, identifying candidate biomarkers that warrant prospective validation.

9.5 Lessons Learned

The most consequential insight concerns multi-seed validation. Early in development, individual runs were used to assess parameter changes, occasionally producing conclusions that did not hold under different seeds. The inherent stochasticity of agent initialization, mutation events, and probabilistic interactions means any single run is a sample from a surprisingly wide outcome distribution. The adoption of ensemble analysis—formalized in the 160-simulation study—was essential for distinguishing genuine parameter effects from noise and producing meaningful confidence intervals.

The decision to incorporate a continuous glucose field, rather than relying solely on agent-based rules, proved a significant source of predictive value. Purely agent-based approaches struggle to represent the spatially heterogeneous metabolic landscape that shapes immune viability. The glucose field introduces emergent spatial structure—depletion zones, nutrient gradients guiding migration—that arises naturally from coupling diffusion physics with agent behavior, and was instrumental in reproducing differential outcomes between well-perfused and poorly perfused tumor regions.

9.6 Closing Remarks

The four objectives stated in Chapter 1—porting to Repast4Py, adding glucose microenvironment sensing, validating against ARON-1 data, and optimizing parameters via Bayesian optimization—have each been addressed. The resulting model integrates metabolic, genetic, hormonal, and pharmacological subsystems into a single agent-based framework reproducing sex-stratified treatment outcomes consistent with clinical observation. While substantial limitations remain, the model establishes an extensible platform for the future directions outlined above and reflects the broader trajectory of computational oncology toward patient-level mechanistic prediction through coupled metabolic, genetic, and immunological dynamics.

A. Parameter Reference Tables

This appendix provides the complete listing of all weight parameters used in the agent-based model, organized by cell type. For a high-level summary, see Chapter 7.

Hormonal Influence Effect Channels

Table A.1: The 17 hormonal influence effect channels through which sex hormones modulate T cell behavior.

Effect	**Description**
KILL_RATE	Cytotoxic kill rate modulation
PROLIFERATION	General proliferation rate
APOPTOSIS	Programmed cell death susceptibility
ACTIVATION	T cell activation state
DIFFERENTIATION	Differentiation probability
TH1_PROLIFERATION	Th1-specific proliferation
TH2_PROLIFERATION	Th2-specific proliferation
IL_4 / IL_6 / IL_13	Interleukin signaling effects
IFN_GAMMA	Interferon-gamma signaling
TNF_ALPHA	Tumor necrosis factor alpha
MHC_I_UPREGULATION	MHC class I expression
MHC_II_UPREGULATION	MHC class II expression
COSTIMULATION_CD28	CD28 costimulatory signal
CTLA4_SUPPRESSION	CTLA-4 checkpoint engagement
PD1_INHIBITION	PD-1 checkpoint engagement

Macrophage and Tumor Cell Weights

Table A.2: Macrophage (M1/M2) and tumor cell weight parameters.

Parameter	**Default**
M1 Macrophage (7 weights + 2 biases)	
w_m1_mutation / b_m1_mutation	1.0 / 0.0
w_m1_move	1.0
w_m1_phagocytosis	1.0
w_m1_digest	1.0
w_m1_t_kill_rate	1.0
w_m1_th1_proliferation	1.0
w_th1_proliferation / b_th1_proliferation	1.0 / 0.0
M2 Macrophage (5 weights + 1 bias)	
w_m2_mutation / b_m2_mutation	1.0 / 0.0
w_m2_move	1.0
w_m2_t_kill_rate	1.0
w_m2_tumor_growth	1.0
w_m2_angiogenesis	1.0
Tumor Cell (8 weights + 4 biases)	
w_tumor_apoptosis_eff / w_tumor_apoptosis_dna	1.0 / 1.0
b_tumor_apoptosis	0.0
w_tumor_growth_eff / w_tumor_growth_dna	0.6 / 1.0
b_tumor_growth	0.0
w_tumor_angiogenesis / b_tumor_angiogenesis	1.0 / 0.0
w_antigen_presentation / b_antigen_presentation	1.0 / 0.0
w_angiogenesis_tumor_growth	1.0
w_gene_pd1_inhibition	1.0

T Cell and Innate Immune Cell Weights

Table A.3: T cell (CD4+, CD8+, Treg) and innate immune cell (mast, NK, pDC) weight parameters.

Parameter	**Default**
CD4+ (10 weights + 3 biases)	
w_cd4_treg_diff_effect / w_cd4_treg_diff_horm	1.0 / 1.0
b_cd4_treg_diff	0.0
w_cd4_th1_proliferation_effect / _horm	1.0 / 1.0
b_cd4_th1_proliferation	0.0
w_cd4_th1_spawn_m1 / w_cd4_th1_spawn_dc	1.0 / 1.0
w_cd4_th2_spawn_m1 / w_cd4_th2_spawn_t	1.0 / 1.0
w_cd4_th2_proliferation_effect / _horm	1.0 / 1.0
b_cd4_th2_proliferation	0.0
Cytotoxic T Cell / CD8+ (5 weights + 2 biases)	
w_cytotoxic_proliferation	1.0
w_cytotoxic_apoptosis / b_cytotoxic_apoptosis	1.0 / 0.0
w_cytotoxic_kill / b_cytotoxic_kill	1.0 / 0.0
w_cytotoxic_pd1_inhibition	1.0
w_cytotoxic_move	1.0
Regulatory T Cell (5 parameters)	
w_treg_t_kill_rate / w_treg_t_proliferation	1.0 / 1.0
w_treg_t_apoptosis / w_treg_activation	1.0 / 1.0
w_treg_dc_phagocytosis	1.0
Mast Cell (6 parameters)	
w_mast_cell_angiogenesis / w_mast_cell_m1_mutation	1.0 / 1.0
w_mast_cell_t_kill_rate / w_mast_cell_tumor_apoptosis	1.0 / 1.0
w_mast_cell_tumor_growth / w_mast_cell_spawn_dc	1.0 / 1.0
NK Cell (2 weights + 1 bias)	
w_natural_killer_kill_rate / b_natural_killer_kill_rate	1.0 / 0.0
w_nkl_t_kill_rate	1.0
Plasmacytoid DC (6 weights + 1 bias)	
w_pdc_nkl_spawn / b_pdc_nkl_spawn	1.0 / 0.0
w_pdc_angiogenesis / w_pdc_treg_diff	1.0 / 1.0
w_pdc_t_proliferation / w_pdc_t_kill	1.0 / 1.0
w_pdc_nkl_kill	1.0

Global and Glucose Weights

Table A.4: Global and glucose weight parameters.

Parameter	**Default**
Global	
w_search_dimension	0.5
w_tumor_growth_threshold	1.0
w_cell_base_death_prob	0.009
w_progressive_exhaustion	0.05
w_ici_effectiveness / w_tki_effectiveness	1.0 / 1.0
w_sex_hormone_cd8	1.0
w_dc_phagocytosis_effect / b_dc_phagocytosis	1.0 / 0.0
w_max_drift	0.25
receptor_threshold_variation	1
Glucose	
w_glucose_diffusion	0.1
w_glucose_decay	0.01
w_glucose_source_rate	5.0
w_glucose_tumor_consumption	2.0
w_glucose_immune_consumption	0.5
w_glucose_growth_sensitivity	1.0
w_glucose_immune_sensitivity	1.0
w_glucose_chemotaxis_strength	0.3

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